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Corporate Adjustment Beyond Leverage

Evidence from European Firms under Earnings-Stripping Rules

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List of abbreviations

BEPS Base Erosion and Profit Shifting

OECD Organization for Economic Co-operation and Development

EU European Union

MNE Multinational Enterprise

ATAD Anti-Tax Avoidance Directive

TCR Thin-Capitalization Rule

ESR Earnings Stripping Rule

EBIT Earnings Before Interest and Taxes

EBITDA Earnings Before Interest, Taxes, Depreciation, and Amortization

GDP Gross Domestic Product

ATT average treatment effect on the treated

PS Propensity Score

SMD Standardized Mean Difference

NACE Nomenclature of Economic Activities in the European Community

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1 Introduction

The deductibility of interest places corporate financing at the intersection of corporate taxation and firm behavior. Because interest payments are generally deductible while returns to equity are not, corporate tax systems create a debt tax shield that lowers the after-tax cost of borrowing and increases the incentive to use debt (Graham, 2003; Mooij, 2011). In multinational groups, the same asymmetry also creates opportunities to place debt where interest deductions are particularly valuable, linking capital structure to the erosion of national tax bases (Huizinga et al., 2008). Policies that restrict interest deductibility therefore address more than an accounting choice. Depending on their design, they may induce firms to relocate or reclassify debt, change total borrowing, substitute toward equity, or adjust other financial and operating policies. Identifying which of these margins responds is essential for understanding the behavioral consequences of such rules.

Early thin-capitalization rules commonly targeted related-party debt through balance-sheet ratios. Although these rules could reduce the targeted internal-debt margin, firms often retained scope to substitute toward external borrowing or to adjust their organizational and reporting structures (Buettner et al., 2012; Wamser, 2014). Modern earnings-stripping rules broaden the restriction by linking interest-deduction capacity to earnings. At the European level, Article 4 of the Anti-Tax Avoidance Directive (ATAD) provides a headline limit of 30% of tax-adjusted Earnings Before Interest, Taxes, Depreciation, and Amortization (EBITDA) for exceeding borrowing costs; Section 2.4 documents the rule and its national options. This design makes the value of the debt tax shield state dependent: the interest generated by a given debt position may be fully deductible when earnings are high but partly restricted when earnings are low. Broader earnings-based rules may therefore affect consolidated financing, yet their behavioral consequences need not stop at the balance sheet. Existing studies report different financing and real responses across countries and policy designs (De Simone et al., 2025; Harju et al., 2025), leaving open which margins adjust when firms become newly exposed to a stricter regime.

Against this background, the central research question is: *Do earnings-stripping rules induce corporate adjustment beyond leverage?* Operationally, the thesis asks how newly exposed firms change debt-to-assets, future earnings volatility, and cash holdings relative to firms already operating under an equally effective limitation. These outcomes represent three potential adjustment margins in response to the same state-dependent tax incentive. First, a lower probability that interest expenses will be deductible reduces debt's expected tax advantage and predicts a decline in leverage. Second, volatile earnings increase the likelihood of entering low-EBITDA states in which a given interest burden becomes constrained, potentially encouraging firms to stabilize subsequent operating performance. Third, cash may become more valuable as a buffer when weak earnings reduce both internal resources and deduction capacity. This last prediction is conditional because firms may instead use cash to repay debt, while deleveraging can itself reduce the need for a liquidity buffer. The three margins can consequently complement or substitute for one another and

need not move together.

The empirical analysis uses an unbalanced 2012–2023 firm-year panel of consolidated Orbis accounts for European non-financial firms. The estimation sample differs across outcomes because each outcome and its controls require different variables and time periods. Treatment is assigned by headquarters country: newly exposed firms are headquartered in countries without a pre-2019 interest-limitation regime classified as equally effective, while comparison firms are located in countries where such a regime already existed. The control group is therefore not unregulated. Firm-level difference-in-differences models compare within-firm changes between the two groups, conditional on firm and year fixed effects and time-varying controls. Leverage and cash use the 2019 implementation benchmark; operational risk instead uses 2017 as a common analytical cutoff for its forward-looking outcome, as detailed in Section 5.1. Event studies and complementary robustness checks examine pre-reform dynamics, observable comparability, alternative controls, pandemic overlap, and heterogeneity in legal and firm-level exposure.

Capital structure provides the clearest result. After the 2019 benchmark, newly exposed firms reduce debt-to-assets by an estimated 2.05 percentage points relative to the comparison group. This corresponds to approximately 9% of the descriptive sample mean and is economically meaningful as well as statistically distinguishable from zero under conventional country-clustered inference. Supporting balance-sheet estimates show higher equity alongside lower current and long-term debt. Their joint direction is consistent with a broader movement away from debt financing rather than only a shift between debt maturities. The debt-to-assets estimate remains negative across the selected robustness checks. The joint pre-trend test for debt-to-assets does not reject the null of no differential pre-reform dynamics, but the marginally significant 2015 and 2016 coefficients and weaker pre-period dynamics for some component outcomes prevent treating the pre-reform paths as fully stable. Capital structure is therefore the most directionally stable empirical margin, although the result remains a conditional relative pattern rather than evidence that excludes every competing country-level explanation.

The evidence beyond financing is less decisive. The estimated relative reduction in adjusted forward earnings volatility is approximately 1.96 percentage points. It has the predicted negative sign and is economically non-trivial, but is imprecisely estimated under conventional inference. The result is consistent with more stable subsequent operating performance among newly exposed firms; it does not directly observe project choice or deliberate managerial risk reduction. Moreover, its dynamic diagnostics and strict pre-pandemic sensitivity check are less supportive than those for leverage. For cash holdings, the estimated relative change is small, negative, and statistically imprecise at -0.39 percentage points. Neither the main specification nor the selected robustness checks provide affirmative evidence of the predicted positive average cash-buffer response. Taken together, the results provide the strongest evidence for financing adjustment, suggestive evidence of lower subsequent earnings volatility, and no affirmative evidence of positive

average cash accumulation.

The contribution of the thesis lies in assessing these three theoretically distinct adjustment margins within a common state-dependent tax-shield framework. Considering them jointly demonstrates why financing, subsequent accounting stability, and liquidity must be evaluated separately rather than inferred from leverage alone. The consolidated financing pattern is consistent with the distinction between broad limits on total net interest and narrower related-party rules that may primarily change debt composition. It also implies that policy evaluations should examine consolidated debt and equity alongside related-party interest flows. The scope of this interpretation is nevertheless limited. Headquarters-country treatment captures exposure to a new policy environment rather than whether the statutory cap bound an individual firm. The estimates should therefore be understood as conditional relative changes for the represented firms with consolidated accounts, not as universal firm-level effects.

The remainder of the thesis proceeds as follows. The literature review covers the theoretical and empirical work on debt bias, thin-capitalization rules, and earnings-stripping rules. The conceptual framework then develops the state-dependent tax-shield mechanism and the three hypotheses. The data and methodology sections describe the sample and empirical strategy. The results section presents the main estimates, and the robustness section examines their sensitivity. A separate limitations section discusses the main limitations concerning data coverage, identification, and statistical inference. The discussion section relates the findings to the existing literature and develops their policy implications, before the final section concludes.

2 Literature Review

2.1 The Debt-Tax Shield: Theoretical Foundations and Economic Significance

A natural starting point for the thin-capitalization and interest-limitation literature is the "debt bias" embedded in most corporate income tax systems: interest is typically deductible, while the return to equity is not (or is treated less favorably). This generates an interest tax shield that lowers the after-tax cost of debt and, all else equal, increases desired leverage. The baseline mechanism matters for the remainder of this review for two reasons. First, it implies that changes in tax parameters should translate into observable financing responses. Second, in the presence of corporate groups and cross-border activity, the same incentive becomes a channel for base erosion, motivating policies that restrict interest deductibility.

Empirically, leverage choices are commonly framed as a trade-off between the tax benefit of debt and non-tax costs such as expected financial distress, agency problems, and external financing constraints. A key implication is that the relevant tax incentive is firm-specific: what matters is whether additional interest deductions are usable at the margin, not merely the statutory rate. Graham (2003) emphasizes this point and motivates the shift from statutory rates to proxies for "marginal tax rates" that incorporate profitability, loss carryforwards, and other features that determine whether the tax shield reduces current tax liability or is deferred.

Across studies, the consensus is that taxes affect capital structure in economically meaningful ways, though effect sizes vary with measurement and context. Feld et al. (2013) summarize the evidence and report a representative semi-elasticity around 0.30: a one percentage-point higher corporate tax rate is associated with about a 0.30 percentage-point higher debt-to-assets ratio. Importantly, responsiveness is stronger when the tax shield is more likely to be usable and when firms can borrow without binding frictions, reinforcing that tax effects operate through both deductibility and financing capacity.

The literature also clarifies the policy motivation. Mooij (2011) highlights that generous interest deductibility can erode the corporate tax base and encourage leverage beyond what is privately optimal, with implications for investment distortions and financial stability. These concerns arise even for purely domestic borrowing, but they become sharper in multinational settings: groups can locate debt where deductions are most valuable and use intra-group interest flows to shift profits across jurisdictions. This observation motivates the next strand of research, which moves from aggregate leverage responses toward the location of debt within multinational groups and the role of internal borrowing in base erosion.

2.2 Multinational Capital Structure: The Transition from External Leverage to Internal Debt Shifting

Debt bias takes on a different character in multinational groups because the key margin is often not how much the firm borrows, but where debt sits within the group. When affiliates face different corporate tax rates, the same borrowing generates different tax savings depending on the jurisdiction in which interest is deducted. Internal capital markets therefore allow Multinational Enterprise (MNE)s to allocate interest-bearing liabilities toward high-tax affiliates and place corresponding interest income in lower-tax parts of the group, reducing the overall tax burden without necessarily changing real activity in the short run.

Early contributions emphasize that this debt-location margin is shaped by both tax incentives and real financing frictions. Desai et al. (2004) show that affiliate leverage reflects local tax conditions as well as the surrounding financing environment, and that internal and external borrowing are not perfect substitutes. A canonical empirical benchmark for tax-motivated debt placement is Huizinga et al. (2008), who relate affiliate leverage to a group-specific incentive to shift debt across jurisdictions: affiliates facing stronger tax differentials within the group carry systematically more liabilities. The central conceptual contribution of this literature is to treat debt as a geographically allocable component of the tax base, responsive to international rate differences rather than only to domestic capital-structure trade-offs.

The same studies also motivate heterogeneity that is important for later policy evaluation. Internal borrowing can serve a genuine funding role when external finance is costly or constrained, implying that restrictions on deductibility may bind more strongly for affiliates that rely on intra-group funds (Egger et al., 2014). In addition, organizational and governance features can limit shifting: partially owned affiliates or ownership conflicts may reduce the feasibility of loading debt into specific subsidiaries even when tax differentials are large (Schindler and Schjelderup, 2012). Cross-country differences in enforcement and institutions further affect the scope for debt shifting and the expected effectiveness of any restriction (Fuest et al., 2011).

These insights map directly into the policy problem that culminates in the Base Erosion and Profit Shifting (BEPS) agenda. Because intra-group lending is scalable and can be organized through standard treasury structures, policymakers increasingly treated interest deductions as a core base-erosion risk. The Organization for Economic Co-operation and Development (OECD)'s BEPS Action 4 report formalizes this concern and develops the policy response documented in Section 2.4. A complementary implication from the broader avoidance literature is that debt shifting is only one margin among several: restricting interest deductions may induce substitution within financing (internal versus external debt) and across profit-shifting tools (e.g., intangibles and transfer pricing), as highlighted in reviews such as Beer et al. (2020) and in empirical work on alternative shifting channels (Dischinger and Riedel, 2011; Dharmapala and Riedel, 2013). Meta-analytical evidence

underscores that estimated magnitudes vary by channel and method, reinforcing the need to study substitution and adjustment margins when evaluating policy (Heckemeyer and Overesch, 2017).

This sets up the next stage of the literature on rule design and effectiveness. Before BEPS and the European Union (EU)-wide push under the ATAD, many countries relied on relatively simple thin-capitalization rules and safe harbors focused on related-party debt. The ensuing empirical question is whether these targeted constraints reduced intra-group borrowing or instead triggered circumvention through relabeling, greater external borrowing, or organizational restructuring, a theme explored in cross-country and reform-based evidence (Weichenrieder and Windischbauer, 2008; Overesch and Wamser, 2010; Buettner et al., 2012).

2.3 First-Generation Thin-Capitalization Rules: Implementation and Empirical Evidence

As research documented that MNEs allocate debt toward high-tax affiliates, many countries adopted first-generation Thin-Capitalization Rules (TCRs) that targeted the related-party channel. The typical administrative approach was ratio-based: rather than measuring "excess" interest directly, deductibility was curtailed when an entity breached a balance-sheet threshold (often a debt-to-equity safe harbor), with a scope focused on intra-group debt. This design aimed to curb base erosion through internal leverage while leaving external borrowing comparatively less constrained.

According to Ruf and Schindler (2015), the effectiveness of anti-avoidance laws is determined by technical design choices, including definitions of debt and the identification of related parties. While narrow rules are intended to target specific profit-shifting behaviors, they often create new opportunities for tax planning. Specifically, firms may bypass these restrictions by replacing internal loans with third-party borrowing, effectively staying within the legal "safe harbor" while achieving the same tax advantages.

Reform-based evidence illustrates both effectiveness and predictable workarounds. Using changes in German thin-cap provisions, Weichenrieder and Windischbauer (2008) show that ratio thresholds can become binding and trigger financing responses, but also encourage threshold management and restructuring. More generally, balance-sheet tests invite adjustments beyond deleveraging: firms can manage reported equity, reroute borrowing across entities, or redesign holding structures to remain compliant.

Cross-country subsidiary evidence supports the view that TCRs reduce the targeted margin when they bind. At the sample-average tax rate, Buettner et al. (2012) estimate that introducing a 2:1 safe-harbor ratio would lower the internal-debt ratio by 3.2 percentage points, or approximately 12% of its mean; when the safe harbor is defined in terms of total debt, the predicted decline is 6.6 percentage points, or 24%. The corresponding decline in total leverage is often smaller, consistent with substitution toward external debt in the German context (Wamser, 2014) and broader financing re-optimization (Overesch and

Wamser, 2010).

These patterns clarify both what first-generation rules achieved and what they did not. If the policy objective is narrowly to reduce manipulable intra-group interest deductions, shifting the composition away from internal debt can be meaningful; if the objective is to reduce the overall debt bias, ratio-based related-party rules may be insufficient. Consistent with this, broader databases suggest that leverage effects depend on whether regimes extend beyond related-party debt and how comprehensively they constrain interest capacity (Blouin et al., 2014). Reviews organize this early evidence around a common conclusion: the key question is not whether TCRs "work" in the abstract, but which margins they move, namely internal debt, total leverage, and restructuring/avoidance (Merlo and Wamser, 2014; Ruf and Schindler, 2015).

This two-sided lesson motivates the policy and research shift that follows. Earnings-based limits change the basis of the restriction from a debt-stock ratio to interest expense relative to earnings and can extend the restriction beyond related-party borrowing. The corresponding literature asks whether these broader rules affect only financing composition or also firm activity.

2.4 From Thin-Capitalization Rules to Earnings-Stripping Rules

The limitations of classic thin-capitalization rules motivated a change in the basis on which deductions were restricted. Whereas first-generation rules typically asked whether related-party debt exceeded a balance-sheet ratio, an earnings-stripping rule can instead restrict the excess of interest expense over interest income relative to an earnings measure. When the restriction covers borrowing costs regardless of whether debt is internal or external, it directly reaches the internal-to-external substitution margin left outside a related-party-only rule.

BEPS Action 4 recommends a fixed-ratio rule that limits net interest deductions to a jurisdiction-specific percentage of tax-adjusted EBITDA within a 10–30 percent corridor. Tax-adjusted EBITDA is calculated by adding net interest expense, depreciation, and amortization back to taxable income under the applicable tax rules. The recommended approach may be supplemented by a group-ratio rule, a de minimis threshold, and provisions carrying forward disallowed interest or unused interest capacity (OECD, 2015, chs. 6–8). In the EU, Article 4 of ATAD 2016/1164 provides that exceeding borrowing costs are deductible only up to 30 percent of the taxpayer's tax-adjusted EBITDA. It also permits Member States to provide a safe-harbor amount, standalone-entity and group-based relief, carryforward or carryback arrangements, and specified exclusions (Council of the European Union, 2016, art. 4(1)–(7)). The Directive therefore establishes a common headline cap while allowing national implementation choices that can affect firms' effective exposure.

Holding exceeding borrowing costs constant and abstracting from these options, lower EBITDA reduces current deduction capacity and higher EBITDA expands it. National options can soften this relationship without removing the profit contingency of the

headline rule. The restriction may consequently become tighter during downturns, a feature emphasized by research on its business-cycle incidence (Ropponen, 2021). This profit-contingent feature provides the institutional basis for the state-dependent tax-shield mechanism developed in Section 3.1.

The broader scope of these rules also changes the relevant policy comparison. Cross-country evidence indicates that restrictions applying to total interest are more likely to reduce consolidated leverage than related-party rules, which often primarily alter the composition of debt (Mooij and Hebous, 2018). Modern evaluations must therefore distinguish genuine deleveraging from debt relocation, liability substitution, reclassification, and reporting responses. These adjustment margins organize the empirical evidence reviewed next.

2.5 Empirical Evidence on Earnings-Stripping Rules

2.5.1 Financing and Leverage Responses

Early evaluations of prototype Earnings Stripping Rules (ESRs), particularly Germany's 2008 interest barrier, document financing responses while also exposing measurement difficulties. Statutory coverage does not map cleanly into economic exposure because earnings capacity, exemptions, escape clauses, and carryforwards jointly determine whether a firm is constrained. Estimated leverage effects therefore depend on how predicted bindingness, anticipation, and relief provisions are modeled. Cutoff-based rules also create incentives to manage interest expense, earnings, and organizational structure around the threshold (Gresik et al., 2017). Evidence from German reforms documents both financing responses conditional on these design features and behavior around deductibility cutoffs and exemptions (Buslei and Simmler, 2012; Dreßler and Scheuering, 2012; Andresen and Thorvaldsen, 2022). Consequently, a post-reform decline in leverage or interest expense may combine genuine debt reduction with debt relocation, liability reclassification, or reporting and timing choices.

Evidence on broad, profit-linked limits comes from both the ATAD reform and earlier national interest barriers. Using the EU-wide implementation, De Simone et al. (2025) report a reduction in debt-to-assets of approximately 2.3 percentage points among treated firms relative to controls. Harju et al. (2025) instead study Finland's national interest barrier, introduced in 2014, using Swedish and Danish MNEs as controls. They estimate that Finnish MNEs reduced financial expenses by 25–30%, but find no statistically significant baseline effect on total, long-term, or short-term debt. Debt declines are detected only among the firms classified as most affected by the reform. Because the data do not separately identify intra-group interest rates and debt, the reduction in financial expenses cannot be interpreted as evidence of general deleveraging. The two studies therefore identify different financing responses: lower debt-to-assets in the ATAD study and lower financial expenses, with more limited debt adjustment, in the Finnish setting.

2.5.2 Real Effects

The modern literature also shows that the effects of ESRs need not stop at the balance sheet. In a cross-country European sample, Leszczyłowska and Meier (2021) estimate that net investment rates fall by 1.9 percentage points under rules restricting total debt and by 3.4 percentage points under rules targeting related-party debt. In the harmonized ATAD setting, De Simone et al. (2025) report a decline in operational risk-taking of approximately 0.7 percentage points, or 13% relative to the treated firms' pre-reform mean, and an investment decline of approximately 11.2 percentage points, or 17.8% of the corresponding pre-reform mean. The study also reports changes in innovation, growth, and financial distress, with larger estimates among firms with high pre-reform operating risk. The Finnish study provides a contrasting result: Harju et al. (2025) find no statistically significant reform effect on EBITDA or turnover during their post-reform period. Thus, their documented reduction in financial expenses was not accompanied by a detectable change in these measures of real activity. The evidence on real effects is therefore broader than investment alone but is not uniform across settings or outcomes.

These findings broaden the policy evaluation from tax planning to real activity, but they do not imply that every observed change is generated by the same adjustment margin. Firms may first use comparatively low-cost reporting, timing, or financing responses and alter operations only when those margins are insufficient. Ownership frictions can restrict the relocation of debt within multinational groups (Schindler and Schjelderup, 2012), while earnings cyclicality changes effective exposure over time (Ropponen, 2021). The availability of relief provisions, internal capital markets, and external finance therefore helps determine whether a firm responds through debt, real activity, or some combination of the two.

3 Conceptual Framework

The institutional architecture documented in Section 2.4 is the starting point for the framework. For the mechanism developed here, the headline cap is represented in stylized form as a limit on current net interest deductions relative to EBITDA. Because deduction capacity varies with operating earnings, the tax value of debt becomes contingent on firm performance: the interest generated by a given debt position may be fully deductible in a high-profit year but partly restricted in a low-profit year. This state dependence may induce adjustment along several margins. The framework focuses on three—capital structure, operational risk, and precautionary liquidity—each with a distinct empirical implication.

3.1 From Debt Tax Shields to State-Dependent Deductibility

Interest deductibility lowers the after-tax cost of debt relative to equity and therefore creates an incentive to use debt financing. The value of this tax shield depends on whether the firm has sufficient taxable income to use the deduction, while non-tax costs prevent leverage from increasing without limit (Graham, 2003; Mooij, 2011). The ATAD rule alters this trade-off by restricting interest deductions through an earnings-based capacity. To isolate this mechanism, let net interest denote the non-negative excess of interest expense over interest income and consider a firm with non-negative EBITDA. Abstracting from the national options documented in Section 2.4, Equation 1 represents the headline relationship in stylized current-period form.

$$\text{Deductible Net Interest}_{it} = \min\{\text{Net Interest}_{it}, 0.30 \times \text{EBITDA}_{it}\} . \tag{1}$$

Correspondingly, Equation 2 gives the amount of net interest restricted in the current period.

$$\text{Restricted Net Interest}_{it} = \max\{0, \text{Net Interest}_{it} - 0.30 \times \text{EBITDA}_{it}\} . \tag{2}$$

These expressions provide a simplified representation rather than a complete statutory test. Safe-harbor amounts, standalone-entity and group provisions, carryforwards or carrybacks, specified exclusions, and other national implementation choices can change whether a particular firm is legally constrained. The simplified relationship nevertheless makes the financing incentive explicit: debt generates net interest expense, but its current tax shield remains fully usable only while that expense fits within the headline earnings-based capacity. Whether the statutory restriction binds a particular sample firm is not observed.

Let τ_{ct} denote the applicable corporate tax rate. Abstracting from the timing value of carryforwards, Equation 3 gives the current tax shield generated by net interest.

$$\text{TS}_{it} = \tau_{ct} \min\{\text{Net Interest}_{it}, 0.30 \times \text{EBITDA}_{it}\} . \tag{3}$$

Because debt-service obligations are relatively fixed in the short run while EBITDA

fluctuates, a decline in earnings can move an unchanged debt position closer to or across the cap. From the perspective of a financing decision, the relevant tax benefit is therefore the expected value of deductions across future earnings states. Firms may adjust before the cap binds, especially when financing and operating policies are costly to reverse. The incentive should be strongest when interest burdens are high, earnings are low or volatile, liquidity is weak, or external finance is limited, while relief provisions and adjustment costs can mute the response. Firms can reduce interest exposure through capital structure, lower the probability of constrained states through operating choices, or buffer their cost through liquidity.

3.2 Capital-Structure Channel

Because debt generates the interest expense subject to the cap, capital structure is the most direct adjustment margin. Lower expected deductibility reduces debt's after-tax advantage and can lead firms to borrow less, repay existing debt, retain earnings, or substitute toward equity (Mooij and Hebous, 2018). The predicted response is therefore a decline in leverage, strongest when interest burdens are high relative to expected deductibility capacity. Empirically, this response must be distinguished from debt relocation, liability reclassification, or reporting responses around the threshold, none of which necessarily represents genuine deleveraging.

3.3 Operational-Risk Channel

The operational-risk channel follows from the profit-contingent nature of the rule. Under the cap, deductibility capacity falls when operating performance weakens. Firms with more volatile earnings face a higher probability of entering low-EBITDA states, and those states increase the probability that interest expenses exceed deductible capacity. Earnings volatility therefore makes the debt tax shield less reliable.

This changes the cost of operating risk. Without an earnings-based interest limit, a bad operating state lowers profits and internal cash flow. With such a limit, the same state may also reduce the firm's ability to deduct interest expenses. The downside state becomes fiscally more costly because profitability and the tax benefit of debt may fall together. Firms may therefore reduce marginal risk-taking by choosing safer projects, slowing volatile expansion, stabilizing cost structures, or avoiding operating strategies with high downside earnings risk (Langenmayr and Lester, 2018; De Simone et al., 2025).

This channel is distinct from deleveraging. Reducing leverage lowers the amount of interest exposure, while reducing operational risk lowers the probability that a given exposure becomes constrained. Operational adjustment may be gradual because firms can alter new projects, cost flexibility, or the pace at which risky activities are scaled without immediately changing their entire business model. The channel should be strongest when pre-reform earnings are volatile, interest burdens are high relative to expected earnings, and

operating choices can be adjusted. Its expected implication is a decline in future earnings volatility.

3.4 Precautionary-Liquidity Channel

The third channel concerns liquidity management. Firms hold cash for transaction needs, precautionary motives, agency considerations, and tax-related reasons (Opler et al., 1999). The precautionary motive is especially relevant when firms face uncertain cash flows or costly external finance: cash allows firms to fund operations, meet obligations, and preserve flexibility when internal resources are weak (Almeida et al., 2004). In an earnings-stripping setting, this motive matters because the rule may bind in precisely the states in which liquidity is most valuable.

Low-profit states create multiple pressures. Operating cash flow is weaker, external finance may be more costly, and debt-service obligations remain. If an EBITDA-based interest limitation also restricts deductibility in those states, the firm loses part of the tax relief that normally accompanies interest payments. Cash can buffer this combined pressure by providing internal resources when profitability is low.

The liquidity response is precautionary rather than mechanical: the rule does not require firms to hold cash, but it can raise its value as insurance when low profits reduce both financing flexibility and tax-shield reliability. Whether observed cash holdings increase is nevertheless ambiguous. Firms may instead use cash to repay debt, deleveraging may reduce the need for a buffer, and cash responds to costs and motives unrelated to the rule (Foley et al., 2007). The predicted sign is therefore positive only if precautionary demand dominates these competing forces; a weak cash response would not invalidate the leverage or operational-risk channels.

The liquidity channel is nevertheless important because it captures a different form of adjustment. Leverage and risk choices reduce the probability or size of a constrained state, whereas cash holdings reduce the cost of reaching such a state. Cash provides option value: it gives the firm time to meet obligations, postpone costly external finance, or adjust debt gradually when profitability is temporarily weak.

3.5 Hypotheses and Testable Implications

The three channels share a common source: the earnings-based cap makes the debt tax shield less reliable in low-profit states. They address different components of exposure. Deleveraging reduces the interest burden that may become constrained; operational-risk reduction lowers the probability of entering a constrained state; and cash holdings reduce the cost of that state if it occurs. The channels can therefore complement one another. They can also substitute when adjustment on one margin reduces the benefit of adjustment on another. In particular, debt repayment may use cash and simultaneously reduce the need for a liquidity buffer.

The framework yields the following hypotheses.

H1 (Capital structure). Firms newly exposed to stricter earnings-stripping rules reduce debt-to-assets relative to firms already operating under an equally effective limitation.

H2 (Operational risk). Firms newly exposed to stricter earnings-stripping rules reduce future earnings volatility relative to firms already operating under an equally effective limitation.

H3 (Precautionary liquidity). Firms newly exposed to stricter earnings-stripping rules increase cash-to-assets relative to firms already operating under an equally effective limitation, provided that the precautionary value of liquidity dominates debt repayment and other uses of cash.

H1 is the most direct prediction because the rule reduces the expected tax benefit of debt. H2 captures real adjustment through the likelihood of future low-EBITDA states. H3 is deliberately conditional because observed cash is an equilibrium outcome of competing uses of funds. The three responses need not occur simultaneously, and the absence of one does not by itself reject the reasoning underlying the others.

4 Data

This chapter describes the Orbis-based firm-year panel used to examine the capital-structure, operational-risk, and liquidity responses of European non-financial firms. It presents the data sources, sample construction, variable definitions, descriptive support, and principal measurement boundaries.

4.1 Data Source and Unit of Observation

The firm-level accounting data come from Orbis, provided by Bureau van Dijk, and the unit of observation is the firm-year. The export contains consolidated accounts, identified by Orbis consolidation codes `C1` and `C2`, so the statements represent the consolidated firm or corporate group rather than an unconsolidated subsidiary. The imported extract spans 2012–2025, but the cleaning procedure retains accounting years 2012–2023; the main implementation-window sample used for the capital-structure and cash-holdings analyses covers 2014–2023. Monetary variables are reported in thousands of US dollars in the Orbis export; logged size measures use values rescaled to millions of US dollars. The panel combines firm identifiers, headquarters country and Nomenclature of Economic Activities in the European Community (NACE) Rev. 2 industry codes with accounting information on assets, sales, debt, equity, earnings, and cash holdings.

The accounting panel is merged with country-year macroeconomic, tax, and institutional data. Annual Gross Domestic Product (GDP) growth and consumer-price inflation come from the World Bank's World Development Indicators, and the combined statutory corporate tax rate comes from the OECD. The long-term interest rate is Eurostat's annual 10-year government-bond yield, expressed in percent. Institutional conditions are measured by the World Bank's Worldwide Governance Indicators for rule of law, regulatory quality, and control of corruption, each expressed on a 0–100 scale (World Bank, 2026a; World Bank, 2026b; OECD, 2026; Eurostat, 2026). Together, these variables characterize the national tax setting, business cycle, financing environment, and institutional conditions surrounding each firm-year.

4.2 Sample Construction and Country Classification

The cleaning procedure restricts the imported panel to 2012–2023, removes financial firms using NACE Rev. 2 codes 64–66, and requires valid firm identifiers, years, and positive total assets. Negative values for accounting levels that should be non-negative—including assets, operating revenue, cash, debt components, and liabilities—are set to missing. Financial expenses and income tax expense are standardized as negative accounting entries, while positive-magnitude copies are retained for constructions that require expense amounts. These rules are applied before the analysis variables are constructed. The descriptive sample-flow and summary tables use a common 2014–2023 implementation-window panel, which provides five accounting years before and five years from the 2019 benchmark

onward for the capital-structure and cash-holdings analyses. The operational-risk analysis instead draws on the broader 2012–2023 constructed panel because it uses a 2017 common analytical cutoff and requires complete forward three-year earnings-volatility windows; each regression ultimately uses its own outcome-specific complete-case sample.

Table 1: Sample restrictions and attrition

Selection step	Observations retained	Firms retained	Countries
Imported Orbis export	71,344	5,096	27
Retain accounting years 2012–2023	61,152	5,096	27
Exclude NACE 64–66 firms	53,844	4,487	27
Require positive total assets	41,750	4,408	27
Restrict to 2014–2023	36,679	4,408	27
Require pre- and post-2019 firm support	33,979	3,602	27
Apply baseline 22-country set	26,212	2,794	22

Notes: Setting implausible levels to missing, harmonizing expense signs, merging country-year variables, and constructing analysis variables do not remove rows and are therefore described in the text rather than shown as sample restrictions. The pre/post support rule requires at least one firm observation on each side of the 2019 cutoff; observations in every year are not required.

Table 1 reports the restrictions that reduce the raw import to the 2014–2023 baseline firm-year panel, together with the observations and firms retained after each step. The final panel contains 26,212 observations from 2,794 firms in 22 countries. The within-firm support requirement is applied before the country restriction, so the final reduction from 33,979 to 26,212 observations reflects the baseline country set alone. The support rule excludes firms observed only before or only after the 2019 cutoff but does not require observations in every year; the resulting panel therefore remains unbalanced.

Treatment status is assigned by headquarters country. Following the European Commission's equally-effective-measures assessment (European Commission, 2018), the control group comprises France, Germany, Greece, Slovakia, Slovenia, and Spain, which had equally effective regimes before 2019; the newly exposed treatment group comprises Belgium, Bulgaria, Croatia, Cyprus, the Czech Republic, Denmark, Estonia, Finland, Hungary, Italy, Lithuania, Luxembourg, Malta, the Netherlands, Portugal, and Sweden. Latvia, Poland, and Romania are excluded because they implemented the earnings-stripping rule in 2018, while Austria and Ireland are excluded because they implemented it in 2021 and 2022, respectively; the United Kingdom is excluded separately to avoid potential distortion associated with its withdrawal from the EU. In the baseline firm-year panel, treated countries contribute 15,514 observations from 1,673 firms, while control countries contribute 10,698 observations from 1,121 firms. Appendix Table 11 documents the country assignments and post-reform rule design.

4.3 Variable Construction

Variables are constructed after the accounting panel has been cleaned and merged with the country-year data. Table 2 summarizes the main outcomes and firm-level controls.

Raw accounting values are not winsorized before ratios are formed. Instead, the resulting continuous analysis variables are winsorized at the pooled 1st and 99th percentiles. Infinite values are set to missing, and logarithms are defined only for strictly positive levels.

If one debt component is missing, the observed component is retained; *DtA* is missing only when both are unavailable. Logged equity, current debt, and long-term debt provide supporting evidence on the balance-sheet adjustment underlying a change in *DtA*.

For the operational-risk analysis, the primary outcome is an adjusted forward earnings-volatility measure constructed for this thesis. It draws on accounting-based risk proxies that capture volatility in scaled profitability relative to a peer benchmark (John et al., 2008; Faccio et al., 2011; Acharya et al., 2011). Related work also uses adjusted forward-three-year constructions (Langenmayr and Lester, 2018; De Simone et al., 2025). For each firm-year, adjusted profitability equals EBIT/assets minus the country–two-digit-NACE–year benchmark calculated in the construction panel. Adjusted *EarnVol* is 100 times the sample standard deviation of adjusted profitability over a complete consecutive forward window from t through $t + 2$. A higher value therefore indicates less stable subsequent operating profitability relative to local peers. The benchmark adjustment is intended to attenuate common variation, but it may also remove responses shared within peer cells. The measure treats positive and negative fluctuations symmetrically and does not directly identify managerial risk choices or whether the statutory restriction binds. Because the windows overlap and the accounting panel ends in 2023, the last observable base year is 2021. Like the other continuous constructed variables, the resulting measure is winsorized at the pooled 1st and 99th percentiles. The unadjusted measure uses the same scaling, sample-standard-deviation convention, and complete forward window without subtracting the peer benchmark.

Table 2: Definitions of main outcomes and firm-level controls

Variable	Construction	Scale
Outcomes		
Debt-to-assets	Long-term debt plus short-term loans and debt, divided by total assets	Ratio
Adjusted *EarnVol*	100 times the complete forward three-year sample standard deviation of EBIT/assets after subtracting the country–two-digit-NACE–year peer benchmark	$100 \times$ SD
Unadjusted *EarnVol*	100 times the complete forward three-year sample standard deviation of unadjusted EBIT/assets	$100 \times$ SD
Cash/assets	Cash and cash equivalents divided by total assets	Ratio
Firm-level controls		
Log assets	Natural logarithm of total assets measured in millions of US dollars	Log level
Log sales	Natural logarithm of operating revenue measured in millions of US dollars	Log level
EBIT/assets	EBIT divided by total assets	Ratio
EBITDA/assets	EBITDA divided by total assets	Ratio
Tangibility	Tangible fixed assets divided by total assets	Ratio
Log sales growth	Change in log operating revenue between consecutive firm-years	Log difference
Country-sector sales growth	Country–two-digit-NACE–year mean of firm-level log sales growth	Log difference

Notes: SD denotes standard deviation. Logs are defined only for strictly positive values. Continuous constructed variables are winsorized at the pooled 1st and 99th percentiles. Complete *EarnVol* windows run from base year t through $t + 2$. Peer benchmarks are leave-one-out in cells with at least two usable observations; singleton cells are benchmarked against their own observation. Country-level controls are described in the text.

The control variables capture differences in size, financing capacity, profitability, collateral, and growth conditions. Log assets measures firm size, which can affect access to external finance and operational diversification. EBIT/assets and EBITDA/assets capture current profitability and internally generated resources, while tangibility reflects the asset base available as collateral. Sales growth is the change in log operating revenue between consecutive firm-years and captures firm-specific expansion; it enters the operational-risk and cash-holdings specifications. The capital-structure specifications instead use the contemporaneous country–two-digit-NACE–year mean of firm sales growth as an accounting-based proxy for the growth environment facing comparable firms. Because this cell mean is contemporaneous and includes the focal firm, it captures realized rather than expected growth opportunities and is not a purely external measure of the firm's opportunity set. Both sales-growth measures serve as controls. Their outcome-specific use and the country-year control sets are specified in Section 5.1.

4.4 Variable Availability and Descriptive Statistics

Table 3 reports availability and descriptive moments in the 2014–2023 baseline firm-year panel. Debt-to-assets and cash/assets are observed for nearly the full panel, whereas

sales growth requires consecutive observations and earnings volatility requires a complete forward three-year window. Part of the risk-sample reduction is therefore mechanically caused by unavailable end-of-window observations, while other outcome-specific losses can arise when a required control is unavailable. This panel provides a descriptive benchmark rather than a universal estimation sample; each empirical block applies its own time window, complete-case requirements, and pre/post-support rule.

Table 3: Main variable availability and descriptive statistics

Role	Variable	N	Missing share	Mean	Median	SD
Capital structure	Debt-to-assets	26,176	0.1%	0.228	0.197	0.199
Risk	Earnings volatility	20,442	22.0%	10.989	4.119	24.823
Risk	Unadjusted earnings volatility	20,442	22.0%	7.636	2.726	15.257
Liquidity	Cash/assets	26,062	0.6%	0.171	0.102	0.195
Growth	Log sales growth	24,551	6.3%	0.069	0.041	0.531
Firm controls	Log assets	26,212	0.0%	5.029	4.963	2.669
Firm controls	EBIT/assets	26,040	0.7%	-0.027	0.037	0.250
Firm controls	EBITDA/assets	25,996	0.8%	0.021	0.072	0.231
Firm controls	Tangibility	26,074	0.5%	0.203	0.117	0.225
Country controls	Corporate tax rate	25,629	2.2%	25.701	25.000	6.281
Country controls	GDP growth	26,212	0.0%	1.744	1.883	3.217
Country controls	Inflation	26,212	0.0%	2.305	1.446	2.919
Country controls	Long-term interest rate	26,115	0.4%	1.251	0.780	1.459
Country controls	Rule of law	26,212	0.0%	79.922	82.473	9.904
Country controls	Regulatory quality	26,212	0.0%	75.509	78.113	7.974
Country controls	Control of corruption	26,212	0.0%	76.356	79.784	14.839

The descriptive moments also reveal differences in the shapes of the outcome distributions. Mean debt-to-assets of 0.228 is moderately above the median of 0.197, while mean cash/assets of 0.171 is substantially above the median of 0.102. The difference is largest for earnings volatility, whose mean of 10.989 exceeds its median of 4.119. These gaps indicate right-skewed distributions, making medians and the robustness of estimates to influential observations more informative than means alone. Treated firms are smaller on average and have lower pre-reform debt-to-assets, profitability, and tangibility, while their cash holdings, sales growth, and earnings volatility are higher. Appendix Table 12 reports these descriptive differences. Treatment and control observations are present in every year from 2014 to 2023, but country contributions are uneven. Sweden is the largest treatment-country contributor, while France and Germany dominate control-country support; Appendix Table 13 reports the complete distribution. The resulting panel provides broad cross-country accounting coverage of firms' financing, operating performance, and liquidity over the ATAD period. The available export does not support market-based risk measures or a direct investment-flow outcome. Because the analysis relies on available consolidated accounts and sufficient time-series support, the estimates apply to firms represented in the outcome-specific estimation samples rather than to all European firms.

5 Methodology

The empirical design compares changes in financial and operating policies among firms newly exposed to ATAD-era earnings-stripping rules with changes among firms in countries whose pre-2019 interest-limitation regimes were classified as equally effective (European Commission, 2018). Section 2.4 distinguishes the OECD policy recommendation from the legal architecture of ATAD Article 4.

Within the retained country sample, treatment is assigned at the headquarters-country level. Equation 4 defines $Treat_c$, which indicates whether country c belongs to the newly exposed group:

$$Treat_c = \begin{cases} 1 & \text{if country } c \text{ lacked a pre-2019 regime classified as equally effective,} \\ 0 & \text{if country } c \text{ had a pre-2019 regime classified as equally effective.} \end{cases} \tag{4}$$

The control group is therefore not an unregulated benchmark. It consists of countries whose pre-2019 interest-limitation regimes were classified as equally effective.

The analysis studies three connected margins of adjustment. The baseline margin is capital structure, measured by debt-to-assets. The second margin is operational risk, measured by forward-looking earnings volatility. The third margin is liquidity, measured by cash holdings. This structure reflects the conceptual framework developed in Section 3: earnings-stripping rules can reduce the tax value of debt directly, but they may also affect real operating risk and precautionary liquidity when interest deductibility becomes more sensitive to profitability.

Because this country-level exposure contrast is not randomly assigned, the analysis identifies differential within-firm changes using the panel design below.

5.1 Difference-in-Differences Framework

The main empirical framework is a firm-level panel difference-in-differences model:

$$Y_{ict} = \beta \left(Treat_c \times Post_t \right) + X'_{ict}\gamma + \mu_i + \lambda_t + \varepsilon_{ict}, \tag{5}$$

where Y_{ict} is the outcome of firm i, headquartered in country c, in year t. $Treat_c$ indicates a headquarters country without a pre-2019 regime classified as equally effective, and $Post_t$ is the outcome-specific policy-period indicator. X_{ict} contains time-varying firm and country controls, μ_i denotes firm fixed effects, and λ_t denotes year fixed effects. The time-invariant treatment indicator is absorbed by firm fixed effects and the common policy-period indicator by year fixed effects, so the coefficient of interest is attached to their interaction. Accordingly, β is the conditional relative change after the relevant benchmark for firms in countries without an equally effective pre-2019 regime, given the included fixed effects, controls, and model sample. It receives a causal interpretation only under the assumptions discussed in Section 5.3. It is not a universal effect of all

interest-limitation rules, an estimate of any single national implementation, or a direct observation of managerial intent.

Capital structure and cash holdings use 2019 as the implementation benchmark. For operational risk, following De Simone et al. (2025), the reform indicator switches to one in 2017. This year is used as a common analytical cutoff rather than as a country-specific implementation date. Because *EarnVol* records performance over t through $t + 2$, an observation based in 2017 spans 2017–2019, while 2016 is the final pre-cutoff base year and the omitted reference year in this thesis.

Firm fixed effects absorb time-invariant firm characteristics, including stable differences in business models, reporting practices, headquarters-country characteristics, and persistent financial policies. Year fixed effects absorb shocks common to all firms in a given year, including Europe-wide macroeconomic and regulatory conditions. They do not absorb country-specific time-varying shocks and do not establish parallel trends; these limitations motivate the controls described next and the diagnostics in Section 5.3.

The firm-level controls differ by outcome because the relevant economic margin differs across leverage, risk, and cash. The sets are chosen to reflect determinants of the relevant dependent variable and the closest feasible outcome-specific specification. The capital-structure specifications control for tangibility, *EBITDA*/*Assets*, log assets, and country-sector sales growth as a proxy for the realized country-sector growth environment. Together, these variables capture collateral, internal financing capacity, size, and the growth environment relevant to debt policy. The operational-risk specifications control for log assets, profitability measured as *EBIT*/*Assets*, firm sales growth, and log sales. This set conditions volatility on balance-sheet scale, current operating performance, recent expansion, and operating scale. The cash specifications control for log assets, profitability measured as *EBIT*/*Assets*, tangibility, debt-to-assets, and firm sales growth. These covariates condition liquidity on size, operating performance, collateral, same-year leverage, and expansion. Across the main specifications, country-level controls include the statutory corporate tax rate, GDP growth, inflation, the long-term interest rate, and institutional indicators for rule of law, regulatory quality, and control of corruption. They account for observed national tax, macroeconomic, financing, and institutional conditions that vary over time. Because the firm controls and their availability differ, the three estimates are conditional on different covariates and may use different samples; they should therefore be interpreted as separate outcome-specific estimates rather than as coefficients from a harmonized system.

Standard errors are clustered by headquarters country because treatment varies at that level, allowing residual correlation within countries across firms and years. The full-control models contain only 20 policy clusters, so conventional country-clustered inference is interpreted cautiously (Cameron and Miller, 2015).

Each model is estimated on its outcome-specific complete-case sample. Within that sample, firms must have at least one observation before and after the relevant cutoff, but

need not be observed in every year; the estimation panels therefore remain unbalanced.

5.2 Outcome-Specific Specifications

The common framework is applied to three outcome blocks that differ in their dependent variables, timing indicators, and firm-level controls.

The capital-structure block applies Equation 5 with $Post2019_t$ and debt-to-assets as the primary outcome. Under H1, a negative β indicates the predicted relative reduction in leverage. Because the ratio can change through movements in debt, equity, or total assets, the same specification is also estimated for logged equity, logged current debt, and logged long-term debt. These component outcomes support the interpretation of the leverage estimate rather than representing separate theoretical channels.

The operational-risk block applies Equation 5 with $Reform2017_t$ and adjusted forward earnings volatility, *EarnVol*, as the primary outcome. Its construction is defined in Section 4 and summarized in Table 2. In the present model, a negative β is consistent with H2 and indicates a relative reduction in subsequent earnings volatility; the outcome is an accounting proxy for the operational-risk channel rather than a direct observation of managerial risk choices. The unadjusted measure uses the same forward window without the peer adjustment and serves as a measurement check.

The cash-holdings block applies Equation 5 with $Post2019_t$ and cash and cash equivalents divided by total assets as the liquidity outcome. Cash holdings capture a precautionary-liquidity margin because liquid assets preserve financing flexibility when internal cash flow is weak and external funds are costly (Opler et al., 1999; Almeida et al., 2004). Under H3, a positive β would indicate the predicted relative accumulation of precautionary liquidity, conditional on the competing uses of cash. This margin is not mechanically the inverse of leverage: a firm can deleverage without accumulating cash, accumulate cash without deleveraging, do both, or do neither. Because the cash model also controls for same-year debt-to-assets, its β is conditional on observed contemporaneous leverage.

5.3 Identification Assumptions and Diagnostics

A causal interpretation of these relative changes requires an outcome-specific counterfactual-trends assumption. Absent the ATAD-era exposure change, treated and control firms would have followed comparable trends in the relevant outcome. This assumption concerns trends rather than pre-reform levels. Treated and control firms do not need to be identical before the reform, but their outcomes must be expected to evolve similarly conditional on fixed effects, controls, and sample restrictions.

The first diagnostic tool is a dynamic event-study specification:

$$Y_{ict} = \sum_{k \neq k_0} \beta_k \left(Treat_c \times \mathbf{1}[t = k] \right) + X'_{ict}\gamma + \mu_i + \lambda_t + \varepsilon_{ict}. \tag{6}$$

The coefficients β_k measure treated-control differences in each year relative to the omitted reference year, conditional on the same outcome-specific controls and fixed effects as the corresponding static model. The omitted reference year k_0 is 2018 for capital structure and cash holdings, and 2016 for operational risk. Coefficients for base years before the relevant cutoff are used to assess whether treated and control firms show differential prior trends. For operational risk, these are pre-cutoff base-year coefficients rather than wholly pre-cutoff outcome windows because some forward windows extend beyond their base year. Coefficients after the relevant cutoff describe the timing of relative changes.

The second diagnostic is an outcome-specific Wald test of the joint null hypothesis that all available pre-cutoff event-study coefficients equal zero relative to the omitted reference year (Wooldridge, 2023; Coube et al., 2026). The test evaluates these coefficients together rather than one at a time and uses the same country-clustered inference as the underlying regression. Its conclusion can therefore differ from the significance of any individual coefficient shown in the event-study plot. Rejection provides evidence of differential pre-cutoff dynamics and weakens a simple causal interpretation; non-rejection does not establish parallel trends because such tests may have limited power against relevant violations (Roth, 2022). The test is therefore interpreted together with the plotted coefficients, sample support, and robustness checks.

Several threats remain: country-level treatment is not randomly assigned; treated and control countries may follow different macroeconomic or institutional paths; ATAD implementation may overlap other reforms; the post-2019 period overlaps the COVID-19 pandemic; and accounting outcomes can reflect reporting or classification choices. These threats motivate the sensitivity analyses described in Section 5.4 and reported in Section 7.

5.4 Comparability and Robustness Strategy

The robustness strategy evaluates five distinct questions: sensitivity to the control set, observable comparability, overlap with the COVID-19 pandemic, heterogeneity in the size of the legal change, and heterogeneity in firm-level exposure. The first three are sensitivity analyses, whereas the final two are mechanism-oriented heterogeneity diagnostics rather than alternative identification strategies. Unless a check requires otherwise, the relevant outcome and timing indicator, firm and year fixed effects, headquarters-country clustering, and within-model pre/post-support rule remain unchanged. The outcome-specific full control sets are retained except when the control set itself is varied, while matching, weighting, and subgroup designs change firms, weights, or country support by construction. Section 7 reports how the results vary across these checks.

5.4.1 Control-Set Sensitivity

Control-set sensitivity asks whether the treatment coefficient depends on the selected time-varying firm and country controls. The capital-structure analysis compares three

specifications: firm controls only; firm controls plus all country controls except the long-term interest rate; and the complete firm-and-country control set. The operational-risk and cash-holdings analyses likewise compare three specifications: firm controls only; firm controls plus macroeconomic and tax controls; and the complete firm-and-country control set, including institutional indicators. Richer country controls can absorb observed national conditions but can also remove country-years or countries when the added variables are unavailable. Differences across specifications may therefore reflect both covariate adjustment and changing model support.

5.4.2 Observable Comparability

Observable-comparability checks ask whether the estimates depend on differences between treated and control firms at an outcome-specific anchor year. For each outcome block, the analysis performs one-to-one Propensity Score (PS) matching without replacement at the outcome-specific anchor year (Caliendo and Kopeinig, 2008; Imbens, 2015). A logit model first estimates treatment propensity from the corresponding firm covariates, after which each treated firm is paired with one control firm whose estimated propensity score differs by no more than 0.001. This maximum admissible distance is the matching caliper. Firms without an eligible counterpart are dropped.

The average treatment effect on the treated (ATT) reweighting design instead retains the anchor pool and assigns the weights in Equation 7:

$$w_i = \begin{cases} 1, & D_i = 1, \\ \widehat{p}_i/(1-\widehat{p}_i), & D_i = 0, \end{cases} \tag{7}$$

where D_i is the treatment status inherited from the firm's headquarters country and $\widehat{p}_i$ is its estimated propensity score (Busso et al., 2014; Imbens, 2004). Control firms that more closely resemble treated firms receive greater weight. This preserves more anchor-year support than discrete matching.

Both designs are used as preprocessing rather than as replacement estimators (Ho et al., 2007). Covariate balance after matching or weighting is evaluated with covariate-level Standardized Mean Differences (SMDs) (Caliendo and Kopeinig, 2008; Imbens, 2015). Match-support and weight diagnostics assess retained comparison support and the concentration of the ATT weights (Busso et al., 2014). The selected or weighted firms are then returned to the full panel, outcome-specific complete cases and pre/post support are reapplied, and the original fixed-effects difference-in-differences and event-study specifications are re-estimated.

5.4.3 COVID-Period Sensitivity

COVID-period sensitivity accounts for the different timing of the outcome measures. Debt-to-assets and cash/assets are annual measures, so the strict timing check removes

observations dated 2020 and 2021. Adjusted $EarnVol_t$, by contrast, is calculated from adjusted EBIT/assets in t, $t+1$, and $t+2$. An operational-risk outcome dated 2018 therefore already contains accounting data from 2020. The strict risk specification consequently retains only observations whose complete three-year window ends before 2020. This leaves 2017 as the only base year at or after the analytical cutoff and provides a deliberately restrictive pre-pandemic-window check with limited post-cutoff support rather than a replacement for the main specification.

The second approach retains the full outcome-specific sample and enters observed national pandemic conditions directly as regression controls. For each outcome, the analysis compares the main specification with variants that add health severity, government stringency, or both controls jointly. Health severity is measured as the logarithm of one plus confirmed deaths per million from World Health Organization reporting distributed by Our World in Data (Mathieu et al., 2020). Government restrictions are measured with the Oxford Stringency Index, scaled in ten-point units (Hale et al., 2021). For capital structure and cash holdings, these controls are measured in the same country-year as the annual outcome. For operational risk, they are aligned with the complete t-to-$t+2$ outcome window: annual deaths per million are summed before the logarithmic transformation, while daily stringency is averaged over all calendar days in the window and then divided by ten. Using only the base-year value would omit pandemic conditions contained in $t+1$ and $t+2$. The frozen Oxford series ends in 2022, so the assigned 2023 stringency value remains a disclosed constructed zero rather than an observed Oxford value.

The strict timing check and the control specifications address different concerns: the first removes pandemic accounting years from the outcome, whereas the second retains them and conditions on measured national pandemic severity and restrictions.

This deliberately simple aggregation provides only a crude measure of national restriction exposure. By reducing each three-year path to a single mean, it does not separately preserve the timing, duration, or intensity of restrictions; distinct exposure paths can yield the same window mean. The resulting specification is therefore interpreted as a crude sensitivity check rather than as a comprehensive adjustment for pandemic exposure.

5.4.4 Pre-Reform Policy Regimes

The policy-regime analysis asks whether estimated responses are concentrated where adoption or tightening represented a larger legal change. Following the Commission's equally-effective classification (European Commission, 2018), treated countries are partitioned into those with no pre-reform interest-limitation rule, those with only a thin-capitalization rule, and those with a pre-existing earnings-stripping rule not classified as equally effective. For each category, the original outcome-specific model is re-estimated using the treated countries in that category and the complete baseline control group. The corresponding outcome-specific controls, fixed effects, clustering convention, complete-case rule, and pre/post-support requirement are reapplied within each subgroup.

The three regime categories are estimated in separate regressions. Each reported test therefore concerns whether that category's coefficient equals zero, not whether the coefficients differ across categories. Because the subgroup models contain only 10–12 country clusters, their patterns are interpreted as suggestive mechanism evidence.

5.4.5 Firm-Level Heterogeneity

Firm-level heterogeneity asks whether the post-2019 leverage response varies with firm characteristics measured before the reform. Unlike in the operational-risk analysis, debt-to-assets remains the outcome; operational risk and accounting pressure serve only as time-invariant classifiers. The full-sample leverage model estimates an average treated-control change; pooled interaction models estimate group-specific responses.

The first classifier, constructed in this thesis from adjusted *EarnVol* observations, measures operational risk before the 2017 analytical cutoff. For each firm, it averages at least two observations with base years from 2012 through 2014. Because each observation spans t through $t + 2$, the latest accounting year entering the classifier is 2016. Classified firms are ranked together within the 20-country leverage sample used in the main analysis and divided into four approximately equal quartiles, from Q1, the lowest-risk group, to Q4, the highest-risk group. The pooled model estimates the post-2019 debt-to-assets response in each quartile and directly tests whether higher-risk firms respond differently from firms in Q1.

The second classifier measures pre-reform accounting pressure relative to the headline threshold documented in Section 2.4. A usable firm-year from 2014 through 2016 requires financial expense, financial revenue, accounting EBITDA, and total assets to be observed, financial revenue to be non-negative, and total assets to be positive. In each such year, the accounting-pressure indicator equals one when the condition in Equation 8 holds:

$$\max\{\textit{Financial Expense} - \textit{Financial Revenue}, 0\} > 0.30 \max\{\textit{Accounting EBITDA}, 0\}. \tag{8}$$

A firm is classified as high pressure when this condition holds in at least half of at least two usable years. A pooled interaction model then compares the post-2019 leverage responses of high- and low-pressure firms. This measure approximates accounting exposure to the headline threshold; it does not reproduce the statutory calculation of exceeding borrowing costs or deduction capacity and cannot show whether the restriction actually binds.

Because constructing the classifiers requires additional historical data, each heterogeneity analysis first re-estimates the average treatment effect among firms that can be classified. The robustness discussion compares this restricted-sample estimate with the corresponding full-sample estimate from the main analysis to disclose whether the additional data requirement changes the average response. The pooled interaction models then estimate responses across exposure groups, while group-specific event studies and joint pre-cutoff tests assess their dynamics and pre-reform patterns.

6 Results

The results follow the three adjustment margins developed in Section 3: capital structure, operational risk, and precautionary liquidity. The coefficient of interest measures the relative change after the outcome-specific benchmark among firms newly exposed to the ATAD-era regime compared with firms in countries that already had an equally effective rule. Each outcome block reports its full-control static difference-in-differences estimate and the corresponding event-study diagnostic.

6.1 Capital-Structure Response

The capital-structure analysis provides the most direct test of the framework. The capital-structure prediction is that firms newly exposed to earnings-stripping rules reduce debt-to-assets relative to firms already subject to an equally effective limitation. The full-control capital-structure specification uses the 2019 implementation benchmark and includes 24,851 firm-year observations from 2,655 firms in 20 countries.

The coefficient does not compare firms with and without any interest limitation. It captures the change in newly exposed countries relative to the contemporaneous change in countries whose pre-2019 rules were classified as equally effective, conditional on firm fixed effects, year fixed effects, and the controls described in Section 5. It should therefore be read as a conditional relative change in leverage around the ATAD-era implementation benchmark, not as a universal effect of earnings-stripping rules.

6.1.1 Average Leverage Effect

Table 4 reports the estimated effect on debt-to-assets and the supporting balance-sheet components. The coefficient on $Treat \times Post2019$ in the debt-to-assets model is -0.0205. Treated firms therefore reduce debt-to-assets by an estimated 2.05 percentage points relative to control firms after the reform benchmark. Relative to the mean debt-to-assets ratio of 22.8 percent reported in Table 3, the point estimate corresponds to a reduction of approximately 9 percent. The estimated response is thus economically meaningful as well as statistically distinguishable from zero under conventional country-clustered inference. The dependent variable is a ratio, so the coefficient is an absolute change in debt relative to assets. Firm fixed effects identify the average within-firm change that remains after common year shocks and observed time-varying covariates are removed. The estimate can consequently combine active debt repayment, less new borrowing, equity growth, and changes in the asset base.

The component estimates indicate that the change in the debt ratio is part of a broader balance-sheet adjustment. Logged equity increases by 0.0510, which corresponds to approximately 5.2 percent. Logged current debt decreases by 0.1045, or approximately 9.9 percent, while logged long-term debt decreases by 0.1536, or approximately 14.2 percent. All three estimates point away from debt financing: the denominator-side equity measure

Table 4: Capital-structure regressions

Model:	Debt/assets (1)	Log equity (2)	Log current debt (3)	Log long-term debt (4)
Variables				
Treat × Post2019	-0.0205***	0.0510***	-0.1045**	-0.1536**
	(0.0068)	(0.0109)	(0.0480)	(0.0652)
Tangibility	0.2114***	-0.5680***	0.3525**	1.691***
	(0.0326)	(0.1190)	(0.1589)	(0.4102)
EBITDA/assets	-0.1096***	0.3395***	-0.8856***	-0.9155***
	(0.0276)	(0.0780)	(0.2092)	(0.1063)
Log assets	0.0118	1.021***	0.8809***	1.015***
	(0.0077)	(0.0303)	(0.0269)	(0.0620)
Country-sector sales growth	-0.0008	-0.0441***	-0.0066	0.0888**
	(0.0030)	(0.0146)	(0.0390)	(0.0357)
Corporate tax rate	-0.0013**	0.0012	-0.0056	-0.0089***
	(0.0005)	(0.0010)	(0.0056)	(0.0029)
GDP growth	-0.0009	0.0021	-0.0095	-0.0015
	(0.0008)	(0.0021)	(0.0055)	(0.0046)
Inflation	0.0017*	-0.0103**	0.0026	0.0097
	(0.0010)	(0.0038)	(0.0146)	(0.0104)
Long-term interest rate	0.0017	0.0106***	0.0485**	-0.0137
	(0.0015)	(0.0036)	(0.0182)	(0.0170)
Rule of law	-0.0031	0.0016	-0.0077	-0.0102
	(0.0021)	(0.0043)	(0.0168)	(0.0192)
Regulatory quality	0.0019*	-0.0031	0.0203**	0.0227**
	(0.0011)	(0.0024)	(0.0085)	(0.0081)
Control of corruption	0.0002	-0.0045*	-0.0059	0.0073
	(0.0008)	(0.0023)	(0.0102)	(0.0057)
Fixed-effects				
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Fit statistics				
Firms	2,655	2,591	2,150	2,215
Years	10	10	10	10
Observations	24,851	23,745	18,634	19,622
Adjusted R^2	0.681	0.979	0.832	0.907
Within R^2	0.041	0.652	0.100	0.197

Notes: The table reports the main full-control specifications with firm and year fixed effects. Standard errors are clustered by headquarters country (20 clusters) and appear in parentheses. The debt/assets coefficient is reported in ratio units; multiplying it by 100 gives the effect in percentage points (pp). Coefficients for the other outcomes are in log points. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

increases while both debt components decrease. This alignment reduces the concern that the debt-to-assets result is driven solely by an increase in total assets or by movement in only one debt category.

The joint decline in current and long-term debt is also informative about the form of adjustment. If firms had responded only by changing debt maturity, the two debt categories would be expected to move in opposite directions. Instead, both coefficients are negative, with a larger proportional estimate for long-term debt. Combined with the positive equity coefficient, this pattern is more consistent with a broad substitution away from debt than with a simple shift between short- and long-term borrowing. It does not identify whether equity issuance, retained earnings, scheduled repayment, or reduced refinancing is the dominant firm-level action, because the accounting data do not separately trace these financing flows.

The component results nevertheless play a supporting role. The component estimates also cannot exclude every accounting reclassification or financing-composition explanation. They are therefore best read as evidence that reinforces the balance-sheet interpretation of the main leverage coefficient, rather than as three additional causal tests.

6.1.2 Dynamic Leverage Pattern

Figure 1 presents the corresponding event-study estimates. The upper-left panel reports debt-to-assets, while the remaining panels show the component outcomes. All coefficients are measured relative to 2018, which is normalized to zero. Appendix Table 14 reports the corresponding coefficients and country-clustered standard errors.

For debt-to-assets, the pre-reform coefficients warrant caution. Relative to the omitted 2018 benchmark, the estimates for 2015 and 2016 are positive and marginally significant at 1.31 and 1.41 percentage points, respectively. The coefficient then narrows to 0.42 percentage points in 2017 before reaching the normalized value of zero in 2018. This pattern raises the possibility that the relative debt ratio was already moving downward before the reform benchmark and should therefore not be characterized as perfectly flat.

At the same time, the joint Wald test does not reject the null hypothesis that the 2014–2017 coefficients are jointly zero relative to the omitted 2018 benchmark (Appendix Table 18). The formal diagnostic thus does not reject a stable pre-reform pattern outright, although this non-rejection neither removes the concern raised by the individual coefficients nor establishes parallel trends.

After the reform benchmark, the debt-to-assets coefficients turn negative. The 2019 estimate is -0.63 percentage points and is imprecisely estimated. The relative decline becomes larger and conventionally precise in 2020, reaching -2.76 percentage points, and remains negative in 2021 at -1.85 percentage points, although the conventional evidence is weaker in that year. The estimates for 2022 and 2023 remain negative but are smaller and statistically imprecise. This pattern is consistent with a financing response that becomes most visible after the implementation year rather than an immediate discrete adjustment in

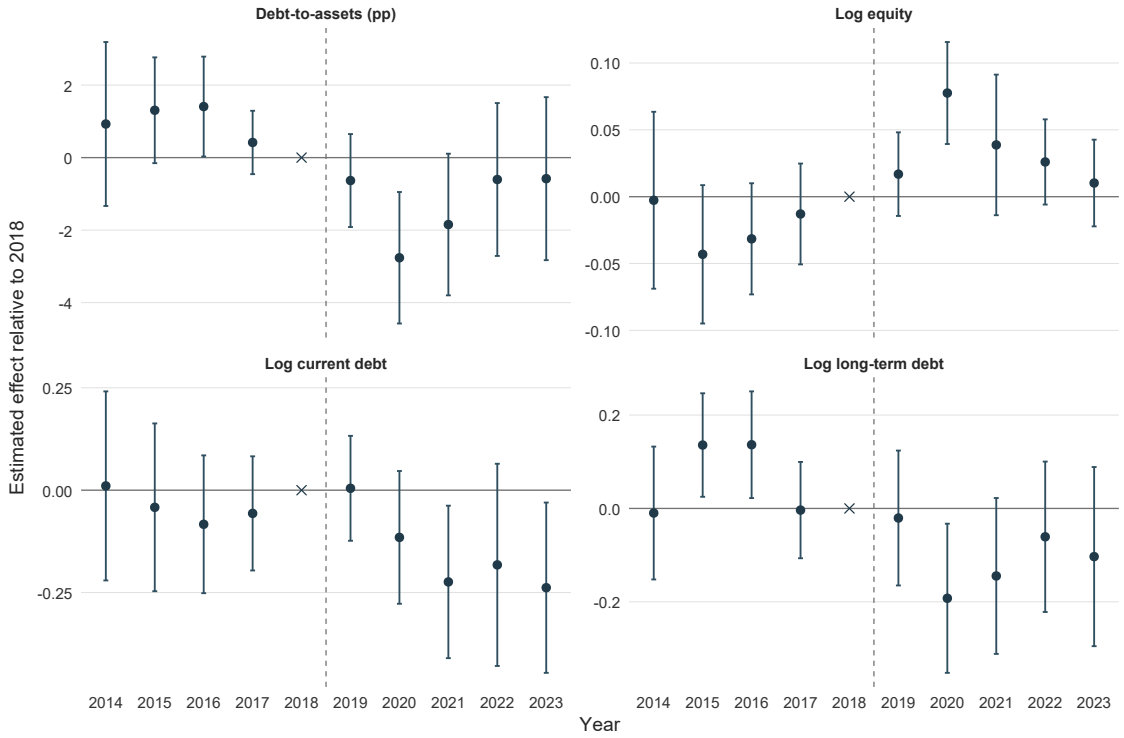


Figure 1: Event-study estimates for capital-structure outcomes

2019. At the same time, the largest coefficient occurs in 2020, when the reform period overlaps with the onset of the COVID-19 pandemic. The event study alone therefore cannot separate reform-related adjustment from differential pandemic exposure.

The dynamic profile also clarifies what the static coefficient summarizes. The average post-2019 estimate of −2.05 percentage points does not imply a constant annual treatment effect. It averages a modest initial difference, a pronounced decline in 2020 and 2021, and smaller later coefficients. Its principal contribution in this chapter is to show that the negative static estimate is accompanied by a visible post-reform movement rather than by a uniformly negative gap throughout the sample.

The concentration of the dynamic decline in 2020 and, to a lesser extent, 2021 raises the possibility that the static result is driven by the pandemic period. The baseline event study cannot resolve that concern. Section 7.1.3 therefore reports the strict timing exclusions and measured-pandemic specifications.

The component panels broadly support the same balance-sheet interpretation. Equity rises most strongly in 2020, while current and long-term debt generally move downward after 2019. Their timing is not uniform, however: the joint tests provide no statistically significant evidence of differential pre-reform dynamics for current debt, weak evidence for equity, and clear evidence for long-term debt (Appendix Table 18). The debt-to-assets diagnostic is more reassuring, but the positive and marginally significant coefficients in 2015 and 2016 leave open the possibility of some pre-existing downward movement toward 2018. The sensitivity of the capital-structure findings to these and related identification concerns is examined in detail in Section 7.1.

6.2 Operational Risk-Taking

The second analysis asks whether exposure to earnings-stripping rules is followed by adjustment beyond financing policy. The operational-risk prediction is a reduction in future earnings volatility because volatile operating performance increases the likelihood that interest expenses become non-deductible in low-profit states. The primary operational-risk outcome is adjusted forward $EarnVol$, defined in Section 4. The relevant interaction is $Treat \times Reform2017$: the indicator switches to one at the 2017 analytical cutoff, while each volatility observation records subsequent performance over t through $t + 2$. The estimation sample contains 18,562 observations from 2,183 firms in 20 countries.

This timing convention changes how the risk coefficients must be read. A 2017 observation summarizes volatility across 2017, 2018, and 2019; a 2018 observation summarizes 2018 through 2020. The indicator marks the analytical cutoff, whereas each coefficient describes volatility over the corresponding forward window. Consequently, a negative coefficient attached to 2017 or 2018 is not, by itself, evidence that firms changed behavior before implementation. It can reflect lower volatility in the later years contained in those forward windows. The complete-window requirement also ends the risk panel in base year 2021, because the 2022 and 2023 windows cannot be observed in data ending in 2023.

The country–two-digit-NACE–year adjustment is intended to attenuate movements common to firms in the same national, sectoral, and year cell. This is particularly relevant in a sample whose post period includes the pandemic. A lower adjusted value indicates that a firm's EBIT/assets path became less volatile relative to its peer benchmark; it does not necessarily mean that the absolute level of EBIT/assets rose or that the measure isolates purely firm-specific risk. The risk analysis therefore concerns the stability of operating performance, not profitability levels or the average scale of operations.

6.2.1 Average Risk Effect

The first row of Table 5 reports an estimated effect of -1.9558 for adjusted $EarnVol$. The estimate implies a reduction of 1.96 percentage points in adjusted earnings volatility among treated firms relative to controls. The corresponding 95 percent confidence interval ranges from -4.11 to 0.19, with a conventional p-value of 0.0722. The point estimate is therefore economically sizeable and has the predicted sign, but it is imprecisely estimated.

For comparison, the same model estimated with unadjusted earnings volatility produces a conventionally precise coefficient of -0.8500. Both measures therefore indicate a negative relative change after the 2017 analytical cutoff. They nevertheless capture different objects: unadjusted earnings volatility retains common country-industry-year movements, whereas adjusted $EarnVol$ measures volatility in firms' deviations from their local benchmark. Because treatment is assigned at the country level, the adjustment may also remove treatment responses shared within treated country-industry cells. Differences in magnitude and precision should therefore be interpreted as a measurement comparison rather than as

Table 5: Baseline earnings-volatility regressions

Model:	Adjusted EarnVol (1)	Unadjusted EarnVol (2)
Variables		
Treat × Reform2017	-1.956*	-0.8500**
	(1.027)	(0.2987)
Log assets	-3.866***	-2.727***
	(0.6234)	(0.2851)
EBIT/assets	-19.07***	-19.42***
	(2.656)	(1.392)
Log sales growth	-0.7505***	-0.2337
	(0.2554)	(0.1465)
Log sales	3.226***	2.242***
	(0.8083)	(0.3558)
Corporate tax rate	-0.0813	-0.0329
	(0.0940)	(0.0347)
GDP growth	-0.0569	-0.0395
	(0.0536)	(0.0321)
Inflation	0.2949	-0.0307
	(0.4445)	(0.1985)
Long-term interest rate	-0.0345	-0.1206*
	(0.1414)	(0.0609)
Rule of law	-0.0373	0.0272
	(0.1770)	(0.0769)
Regulatory quality	0.0211	0.0011
	(0.1178)	(0.0656)
Control of corruption	0.0799	0.0352
	(0.1157)	(0.0544)
Fixed-effects		
Firm	Yes	Yes
Year	Yes	Yes
Fit statistics		
Firms	2,183	2,183
Years	9	9
Observations	18,562	18,562
Adjusted R^2	0.492	0.593
Within R^2	0.042	0.102

Notes: The table reports the main full-control specifications with firm and year fixed effects. Standard errors are clustered by headquarters country (20 clusters) and appear in parentheses. Both outcomes equal 100 times the forward-looking three-year standard deviation of EBIT/assets, so coefficients are expressed in percentage points (pp). Adjusted EarnVol removes the country-industry-year benchmark before volatility is calculated. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

evidence that either specification captures a stronger response.

6.2.2 Dynamic Risk Pattern

Figure 2 plots the dynamic coefficients for both volatility measures relative to the omitted base year 2016. Appendix Table 15 reports the corresponding coefficients and country-clustered standard errors. Although the confidence intervals for the individual adjusted pre-cutoff coefficients include zero, the joint test of the 2013–2015 coefficients yields $p = 0.0836$, providing weak evidence that they differ collectively from the omitted 2016 base year (Appendix Table 18). For unadjusted $EarnVol$, the corresponding joint test yields $p = 0.0039$, even though only the 2015 coefficient is individually marginally significant. The joint tests account for covariance among the coefficients and therefore need not match the significance of any single plotted estimate. These diagnostics weaken the case for a stable pre-cutoff pattern, particularly for the unadjusted measure.

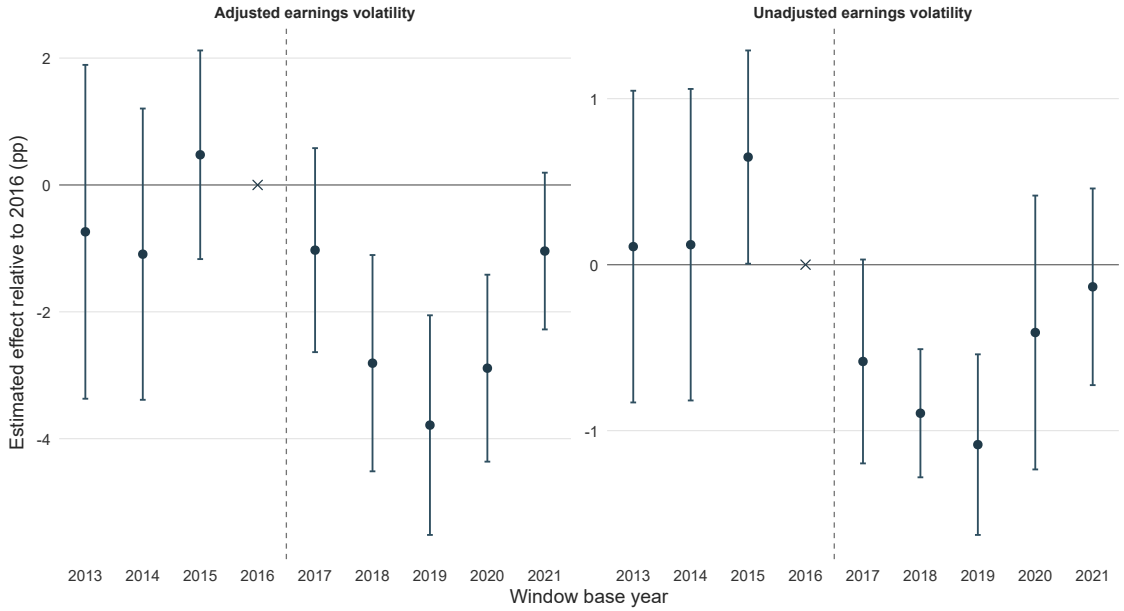


Figure 2: Event-study estimates for earnings volatility

The adjusted coefficients for base years on or after the 2017 cutoff are consistently negative. The estimate is −1.03 in 2017 and is imprecisely estimated. It becomes −2.81 in 2018, −3.79 in 2019, and −2.89 in 2020, with all three estimates conventionally distinguishable from zero. By 2021, the estimate has narrowed to −1.04 and is again imprecise. Because each coefficient describes a forward-looking volatility window, these base-year movements should not be read as year-specific changes in contemporaneous operating risk. Instead, they show that windows spanning the implementation period are associated with lower subsequent volatility among treated firms.

The concentration of the largest estimates in the 2018–2020 base years is consistent with this window structure. These observations contain operating outcomes from 2018 through 2022 and therefore load heavily on the implementation and pandemic periods. The

smaller 2021 estimate is based on the final observable window, 2021–2023. Its attenuation may indicate that the relative volatility difference was temporary, but it may also reflect changing support at the end of the panel or the normalization of extraordinary pandemic-era variation. The dynamic evidence is informative about timing, yet it cannot uniquely assign the lower volatility to deliberate project selection rather than to other operating adjustments or differential shocks.

The post-cutoff pattern is directionally consistent with operational adjustment: firms facing more state-dependent interest deductibility appear to experience less volatile future operating performance. The identification evidence is nevertheless weaker than in the leverage analysis. The static estimate for adjusted forward $EarnVol$ receives only marginal conventional support, and the adjusted event study has a borderline pre-trend diagnostic. The results are therefore consistent with the operational-risk prediction, but they do not establish a decisive causal mechanism linking the reform to firms' underlying project or production choices.

6.3 Cash Holdings

The final outcome tests the precautionary-liquidity channel. The prediction is that treated firms increase cash relative to assets if the value of a buffer against low-profit states dominates debt repayment and other uses of liquidity. The full-control cash specification instead produces a small, negative, and statistically imprecise estimate of −0.39 percentage points ($p = 0.5720$). It therefore provides no evidence of the predicted positive average cash-buffer response. The specification uses $Treat \times Post2019$ and contains 22,570 observations from 2,427 firms in 20 countries.

Cash/assets measures the stock of immediately observable liquidity relative to the balance sheet. It does not record the gross inflows and outflows that produce that stock. A firm could generate additional internal funds and use them immediately to repay debt, leaving the cash ratio unchanged; another could raise debt and hold the proceeds as cash, increasing cash without improving net liquidity. The precautionary-liquidity prediction is therefore tested against a net accounting outcome shaped by several policies. This makes the cash prediction less mechanical than the capital-structure prediction and places particular weight on the direction and precision of the estimated average response.

6.3.1 Average Cash Effect

Table 16 in the Appendix reports the complete regression results. The coefficient of −0.0039 for cash/assets corresponds to a relative decrease of 0.39 percentage points, with a 95 percent confidence interval from −1.81 to 1.03 percentage points. The point estimate is small relative to the mean cash/assets ratio of 17.1 percent in Table 3, while the confidence interval remains compatible with modest changes in either direction. The result therefore neither supports the predicted increase in precautionary cash holdings nor establishes that

exposure to the reform caused cash holdings to fall.

6.3.2 Dynamic Cash Pattern

Figure 3 in the Appendix presents the event-study coefficients relative to 2018. Appendix Table 17 reports the corresponding coefficients and country-clustered standard errors. The joint test provides no statistically significant evidence of differential pre-reform dynamics (Appendix Table 18). The individually imprecise pre-period estimates also show no clear monotonic pattern.

The post-reform path does not show the positive shift implied by precautionary accumulation. The estimates from 2019 through 2023 are -0.29, -0.98, -0.50, -0.80, and -1.29 percentage points. All are negative but imprecisely estimated. The dynamic results therefore provide no affirmative support for a broad increase in cash holdings, but they also do not establish a systematic decline.

This informative non-result narrows the interpretation of firms' response: newly exposed firms do not appear to build larger average cash buffers, at least when liquidity is measured as cash and cash equivalents relative to total assets. COVID-period sensitivity and the other selected robustness checks are reported in Section 7.

7 Robustness

The baseline results differ in strength across the three outcomes. The capital-structure analysis shows a negative and directionally stable debt-to-assets response; the operational-risk analysis provides suggestive evidence of lower earnings volatility but raises greater dynamic-identification concerns; and the cash analysis does not support the predicted positive liquidity response. This chapter examines whether that hierarchy survives five selected challenges: variation in the control set, observable-comparability preprocessing through matching and reweighting, the overlap with the COVID-19 pandemic, heterogeneity in countries' pre-reform policy regimes, and differences in firm-level exposure.

Section 5.4 describes how these checks are implemented and which elements of the main specifications remain unchanged. Throughout this chapter, the main specification for each outcome is the corresponding specification reported in Section 6. The present chapter reports how its conclusions vary across the selected checks.

7.1 Capital-Structure Robustness

7.1.1 Control-Set Sensitivity

The debt-to-assets estimate from the main specification is a relative reduction of 2.05 percentage points (country-clustered standard error: 0.68 percentage points). Panel A of Table 6 compares three specifications. The estimate remains negative and conventionally significant throughout, ranging from -2.78 percentage points with firm controls only to -1.93 percentage points when all controls from the main specification except the long-term interest rate are included. The full-control estimate lies inside this range.

The specifications also differ in model support. The firm-only model retains all 22 countries. Requiring the corporate-tax rate excludes Cyprus, while additionally requiring the long-term interest rate excludes Estonia, reducing support to 21 and 20 countries, respectively. The comparisons therefore combine changes in controls with changes in sample composition. The narrower control sets trade some adjustment for observed country-level conditions for broader country and firm coverage, so the exercise is best read as a joint sensitivity check rather than as a ranking of the individual controls. Across all three control sets, logged equity remains positive and logged long-term debt remains negative and conventionally significant. Logged current debt is also negative throughout, although its precision varies. The broad movement toward higher equity and lower debt is therefore directionally stable across the three specifications, subject to the accompanying changes in model support.

Table 6: Treatment-effect sensitivity across control sets

Outcome	Control set	Effect	SE	p-value	Observations	Firms	Countries
Panel A: Debt-to-assets (percentage points)							
	Firm controls only	-2.78**	1.13	0.0223	25,527	2,728	22
	No interest-rate control	-1.93**	0.70	0.0118	25,011	2,672	21
	Full usable controls	-2.05***	0.68	0.0068	24,851	2,655	20
Panel B: Adjusted earnings volatility							
	Firm controls only	-1.67	1.00	0.1085	19,083	2,244	22
	Firm plus macro/tax controls	-2.06**	0.97	0.0477	18,562	2,183	20
	Full usable controls	-1.96*	1.03	0.0722	18,562	2,183	20
Panel C: Cash/assets (percentage points)							
	Firm controls only	-0.53	0.96	0.5843	23,172	2,491	22
	Firm plus macro/tax controls	-0.30	0.71	0.6798	22,570	2,427	20
	Full usable controls	-0.39	0.68	0.5720	22,570	2,427	20

Notes: SE denotes standard error. Every specification includes firm and year fixed effects and imposes outcome-specific complete-case and firm-level pre/post-support requirements. Standard errors are clustered by headquarters country. Effects for debt-to-assets and cash/assets are reported in percentage points. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Appendix Tables 19, 20, and 21 report the complete leverage, operational-risk, and cash-holdings regressions underlying the control-set comparison.

7.1.2 Observable Comparability

The selected 2018 propensity-score match and ATT reweighting described in Section 5.4 both rebuild the leverage comparison around firms with more similar observed characteristics. Panel A of Table 7 shows that both designs materially improve observed balance. The maximum absolute standardized mean difference falls from 0.384 in the unweighted 2018 pool to 0.077 after matching and 0.063 under ATT reweighting. Matching retains 1,920 of the 2,723 anchor-year firms, whereas reweighting retains the full pool with a maximum control weight of 6.33. Panel B reports the corresponding treatment estimates. The matched regression produces a 1.43-percentage-point reduction in debt-to-assets and the weighted model a 2.19-percentage-point reduction. Both are conventionally significant, although the matched estimate is smaller in magnitude than the main-specification estimate.

The dynamic diagnostics are particularly important because balance in observed levels does not ensure comparable outcome trends. The joint tests for the 2014–2017 debt-to-assets coefficients relative to 2018 yield p-values of 0.3700 after matching and 0.3645 under ATT reweighting. Neither test rejects, and both are at least as reassuring as the corresponding unmatched test for the main specification. This conclusion applies to the primary ratio outcome rather than every component: the long-term-debt pre-trend test rejects in both designs, and the weighted equity test also rejects. The component coefficients remain directionally coherent—higher equity and lower current and long-term debt—but their dynamics prevent treating them as equally strong identification evidence.

Table 7: Matching and ATT-reweighting estimates and diagnostics

Panel A: Design and observed balance

Outcome block	Anchor pool	Matched firms	Max. SMD: raw	Max. SMD: match	Max. SMD: ATT	Max. ATT weight
Leverage	2,723	1,920	0.384	0.077	0.063	6.33
Operational risk	2,229	1,626	0.325	0.065	0.029	5.60
Cash holdings	2,470	1,718	0.360	0.112	0.050	7.13

Panel B: Primary treatment estimates

	Main model		PS match		ATT	
	Estimate	SE	Estimate	SE	Estimate	SE
Debt-to-assets (pp)	-2.05***	0.68	-1.43**	0.66	-2.19***	0.60
Adjusted EarnVol	-1.96*	1.03	-1.85*	1.06	-1.40	0.89
Cash/assets (pp)	-0.39	0.68	0.00	0.71	-0.41	0.70

Panel C: Joint pre-trend test p-values

	Main model	PS match	ATT
Debt-to-assets	0.1959	0.3700	0.3645
Adjusted EarnVol	0.0836	0.0034	0.0265
Cash/assets	0.3783	0.0810	0.0614

Notes: SE denotes standard error. "Main model" denotes the corresponding unmatched specification reported in Section 6. The selected propensity-score design is 1:1 nearest-neighbor matching without replacement and with a 0.001 caliper. ATT weights equal one for treated firms and $\widehat{p}_i/(1-\widehat{p}_i)$ for controls. Leverage and cash use 2018 anchor samples; operational risk uses 2016. The balance statistic is the maximum absolute standardized mean difference across the matching covariates. Regression samples again impose outcome-specific complete cases and firm-level pre/post support. Effects for ratio outcomes are in percentage points. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Panel C of Table 7 reports the corresponding joint pre-trend tests. Appendix Tables 22, 23, and 24 report, for the three primary outcomes, the complete static regressions under the main, matched, and ATT-weighted specifications.

7.1.3 COVID-Period Sensitivity

The post-2019 leverage period overlaps with the pandemic, and the baseline event study shows its largest decline in 2020. Table 8 therefore presents two complementary checks. The strict timing model excludes both 2020 and 2021. Its coefficient is -1.50 percentage points, compared with -2.05 in the full sample, and remains conventionally significant with a p-value of 0.0288. The attenuation indicates that the pandemic years contribute to the magnitude of the full-sample estimate, but those years do not solely generate the negative estimate.

The measured-pandemic specification adds log confirmed deaths per million and the Oxford Stringency Index jointly, as described in Section 5.4. The resulting leverage estimate is -1.87 percentage points and remains significant below the 1 percent level. The estimate from a deaths-only specification is almost identical to the main-specification estimate; the modest attenuation primarily appears when stringency is included.

Table 8: COVID-period sensitivity: timing restrictions and pandemic controls

Outcome	Main model	Strict timing check	Deaths and Oxford jointly
Debt-to-assets (pp)	-2.05***	-1.50**	-1.87***
	(0.68)	(0.63)	(0.63)
Adjusted EarnVol	-1.96*	-0.54	-2.13*
	(1.03)	(0.92)	(1.14)
Cash/assets (pp)	-0.39	-0.42	-0.25
	(0.68)	(0.58)	(0.63)

Notes: Country-clustered standard errors appear in parentheses. The strict timing check excludes 2020–2021 for debt-to-assets and cash/assets. For operational risk, it retains only forward volatility windows satisfying $t + 2 < 2020$. The joint model includes log confirmed deaths per million and the Oxford Stringency Index. Risk exposures are aggregated over t to $t+2$. Firm and year fixed effects are retained. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Appendix Tables 26, 27, and 28 report the complete outcome-specific regressions for the main, strict-timing, deaths-only, Oxford-stringency-only, and joint specifications.

7.1.4 Pre-Reform Policy Regimes

A further check asks whether the response differs across pre-reform policy regimes. Table 9 reports the three regime comparisons defined in Section 5.4: no prior interest-limitation rule, only a thin-capitalization rule, and a pre-existing but non-equivalent earnings-stripping rule.

The first column of Table 9 shows a 3.01-percentage-point decline in debt-to-assets for the no-prior-rule group. The thin-cap and pre-existing-ESR estimates are considerably smaller and imprecise. The largest point estimate therefore occurs in the no-prior-rule group. This pattern is suggestive rather than conclusive: each regime is estimated in a separate regression with only 10–12 country clusters, and the reported p-values test each coefficient against zero rather than equality across regimes. Differences between rows should consequently not be treated as precisely estimated cross-regime contrasts.

Table 9: Treatment effects by pre-reform policy regime

	Debt-to-assets (pp)		Adjusted EarnVol		Cash/assets (pp)		Clusters
Pre-reform regime	Effect	SE	Effect	SE	Effect	SE	
No prior interest-limitation rule	-3.01***	0.26	-3.77*	1.81	-0.17	0.75	10
Thin-capitalization rule only	-0.82	1.12	0.33	0.50	0.31	0.66	12
Non-equivalent earnings-stripping rule	-0.55	0.89	-0.90	1.42	0.18	0.66	10

Notes: SE denotes standard error. Each row is based on separate outcome-specific regressions that retain all baseline control countries and one treated-country subgroup. Standard errors are clustered by headquarters country. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Appendix Tables 29, 30, and 31 report the complete regressions for the three pre-reform policy regimes.

7.1.5 Firm-Level Heterogeneity in Leverage

Firm-level heterogeneity asks whether leverage adjustment differs across firms with different pre-reform exposure. Debt-to-assets remains the outcome. Firms are classified either by pre-cutoff operational risk or by pre-reform accounting pressure, as detailed in Section 5.4.5. The latter is based on consolidated financial accounts and only approximates exposure to the 30-percent threshold; it does not show whether the statutory restriction actually bound a firm.

Because both classifiers require additional historical data, the main leverage specification is first re-estimated separately in each classifier-specific sample. In the operational-risk-classified sample, this restricted-sample estimate attenuates from the full-sample estimate of -2.05 to -1.39 percentage points and becomes less precise ($p = 0.094$). In the accounting-pressure-classified sample, the corresponding re-estimate remains close to the full-sample estimate, at -2.00 percentage points ($p = 0.033$). These comparisons show how the additional data requirements change the average estimate before differences between exposure groups are considered; they are not tests of coefficient differences across samples.

Table 10: Firm-level heterogeneity in the debt-to-assets response

Group	Effect (pp)	SE (pp)	p-value
Panel A: Debt-to-assets effects by pre-cutoff operational-risk quartile			
Q1, lowest risk	0.31	1.25	0.808
Q2	-1.33	0.96	0.182
Q3	-1.70	1.65	0.315
Q4, highest risk	-3.06***	0.82	0.001
Panel B: Debt-to-assets effects by pre-reform 30-percent accounting pressure			
Low pressure	-1.88**	0.81	0.031
High pressure	-1.46	0.96	0.145

Notes: SE denotes standard error. The outcome is debt-to-assets. Estimates are in percentage points and come from pooled interaction models with the full leverage control set, firm and year fixed effects, and standard errors clustered by headquarters country. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Panel A of Table 10 shows a pronounced operational-risk gradient. The estimated debt-to-assets response is approximately zero in Q1 and becomes increasingly negative through Q4, where it reaches -3.06 percentage points and is conventionally significant at the 1 percent level. This pattern is consistent with unstable earnings capacity strengthening the incentive to reduce debt. Its magnitude should not be compared mechanically with the full-sample estimate: restricting the analysis to firms with sufficient pre-cutoff risk data already attenuates the average estimate by 0.66 percentage points, from -2.05 to -1.39. The heterogeneity evidence nevertheless remains suggestive because Q4 control firms come from only four countries.

Panel B does not display the predicted pattern for the accounting-pressure proxy. The estimated effect is -1.88 percentage points for low-pressure firms and -1.46 percentage points for high-pressure firms. To determine whether the responses significantly differ

across the two groups, the relevant statistical comparison is the difference between their estimated effects rather than their individual significance levels. This high-minus-low contrast is +0.42 percentage points, with a p-value of 0.664, so the null hypothesis of equal effects cannot be rejected. The proxy-defined groups therefore provide no evidence of stronger deleveraging among high-pressure firms. Failure to reject equality across these groups does not establish that firms actually constrained by the rule responded no more strongly, because consolidated accounting values do not reproduce taxable borrowing costs, tax-adjusted EBITDA, carryforwards, relief provisions, or the relevant taxpayer-level calculation. The proxy may therefore classify actual exposure imperfectly in both directions. Appendix Table 32 reports the complete pooled interaction regressions underlying these estimates.

7.1.6 Capital-Structure Synthesis

Taken together, the debt-to-assets response remains negative across control sets, matching, ATT reweighting, pandemic adjustments, legal-regime splits, and firm-level exposure groups. Matching and weighting materially improve observable balance, and their primary-outcome pre-trend tests do not reject. The pandemic years contribute to the magnitude but do not fully account for it, while the largest estimates occur in the no-prior-rule group and among firms with high pre-cutoff operational risk. The accounting-pressure split does not reveal the same gradient. Several checks also rely on smaller samples or fewer country clusters than the main specification. Their estimates are therefore less precise. Capital structure is nevertheless the most directionally stable empirical margin, but unobserved country-specific shocks, concentrated subgroup support, and weaker component dynamics materially qualify the strength of that conclusion.

7.2 Operational-Risk Robustness

The adjusted-$EarnVol$ estimate from the main specification is -1.96, with a country-clustered standard error of 1.03 and a p-value of 0.0722. Because both $EarnVol$ measures are scaled as 100 times a forward three-year standard deviation of EBIT/assets, with peer adjustment applied to the adjusted measure, all coefficients in this subsection are expressed in percentage points. Panel B of Table 6 shows that the adjusted estimate's sign is stable across control sets, but its precision is not. The firm-only estimate is -1.67 and statistically imprecise; adding macroeconomic and tax-rate controls produces -2.06, significant at 5 percent; and adding the institutional controls yields the main-specification estimate of -1.96, significant only at 10 percent. The unadjusted measure is negative and conventionally significant in every control set, with estimates between -0.95 and -0.85. This stronger static precision does not provide stronger identification evidence: the joint test of the unadjusted measure's 2013–2015 coefficients relative to the omitted 2016 base year rejects with $p = 0.0039$, indicating differential pre-cutoff dynamics.

The matching and weighting results sharpen this caution. The 2016 anchor designs reduce the maximum absolute standardized mean difference from 0.325 to 0.065 under matching and 0.029 under ATT reweighting. Despite the improvement in observed balance, the adjusted-risk estimate attenuates to −1.85 in the matched sample and −1.40 under reweighting; the former is significant only at 10 percent, and the latter is not conventionally significant. More importantly, the matched and weighted joint pre-trend tests reject with p-values of 0.0034 and 0.0265 (Appendix Table 25). Observable balance in the anchor-year characteristics therefore does not resolve differential pre-cutoff base-year dynamics.

The outcome's forward construction makes pandemic timing especially consequential. The strict model in Table 8 retains only volatility windows whose final year precedes 2020. Its estimate falls to −0.54 and becomes imprecise. This sample contains only one base year at or after the 2017 analytical cutoff, 2017, so it is a low-power diagnostic rather than a replacement specification. By contrast, aligning deaths and stringency with the complete t to $t + 2$ window leaves the full-sample coefficient at −2.13, close to the main-specification estimate, though still imprecise at conventional levels. The national controls thus do not explain the negative full-sample coefficient.

The regime results in Table 9 are partly consistent with the mechanism. The no-prior-rule group has the largest negative estimate, −3.77, which is significant only at 10 percent. The pre-existing-ESR estimate is negative but imprecise, while the thin-cap estimate is small and positive. The largest adjustment therefore appears where the legal shock was greatest, as in the leverage analysis, but the full cross-regime pattern is mixed.

Overall, the selected checks preserve a predominantly negative direction for adjusted earnings volatility but weaken a decisive causal mechanism claim. Precision changes with the control set, the ATT estimate attenuates, matched and weighted pre-trend tests reject, and the strict pre-pandemic-window model provides little affirmative evidence. The operational-risk result should therefore remain characterized as suggestive and subject to material identification qualifications.

7.3 Cash-Holdings Robustness

Across all selected robustness checks, no specification reveals a positive and statistically precise average cash-holdings response. This consistency reinforces the baseline conclusion that the data do not support precautionary cash accumulation following the reform. Panel C of Table 6 shows estimates of −0.53, −0.30, and −0.39 percentage points across the firm-only, macro/tax, and full-control specifications. All are small, negative, and statistically imprecise, so control selection and the associated expansion from 20 to 22 countries do not alter the conclusion.

Observable-comparability preprocessing yields the same substantive conclusion. Matching reduces the maximum absolute standardized mean difference from 0.360 to 0.112, while ATT reweighting reduces it to 0.050. The matching result is a substantial improvement. The matched cash coefficient is effectively zero and the weighted estimate is −0.41 percentage

points. Neither is precise. These models are consequently stress tests rather than superior cash specifications.

The pandemic and regime checks likewise fail to reveal a positive average response. Excluding 2020–2021 leaves a −0.42-percentage-point estimate, while adding deaths and stringency yields −0.25 percentage points; both are imprecise. Across pre-reform regimes, the estimates range only from −0.17 to 0.31 percentage points and none is statistically distinguishable from zero. In particular, the countries with no prior interest-limitation rule do not exhibit hidden precautionary accumulation. Taken together, the selected control, comparability, pandemic, and legal-exposure checks leave the cash analysis as an informative non-result.

8 Limitations

The empirical design provides a structured comparison of firms newly exposed to the ATAD-era regime with firms in countries that already had an equally effective interest-limitation rule. Firm and year fixed effects, country-level controls, event studies, and the robustness analyses in Section 7 all strengthen this comparison. The findings nevertheless remain conditional on the available data, the treatment definition, and the assumptions of the research design. These qualifications do not affect all results equally. They concern two distinct questions: how far the estimates generalize beyond the observed sample and whether the relative changes can be attributed to the reform rather than to other differences between treated and control countries. Within those boundaries, the relative decline in leverage is the most consistent finding, the operational-risk evidence is more tentative, and the cash estimate remains inconclusive about changes in precautionary liquidity.

8.1 Data and Sample Coverage

The Orbis panel offers harmonized accounting information for a large set of firms across several European countries. To maintain comparability, the analysis focuses on non-financial firms with consolidated accounts and sufficient information for the relevant models. The construction and resulting sample are documented in Section 4 and Table 1. This focus means that the estimates apply most directly to the represented consolidated firms, which may be larger and more organizationally complex. The main consequence is therefore one of external validity: the estimated average should not automatically be interpreted as the response of every European firm. This sample definition does not by itself bias the within-firm coefficients. Such bias would require coverage or reporting changes to be systematically related to both treatment status and the outcomes, for which the present analysis provides no direct evidence.

Requiring usable observations before and after the relevant cutoff ensures that each coefficient is based on within-firm changes over the policy period. The trade-off is that firms observed on only one side of the cutoff do not contribute, including post-cutoff entrants and firms that disappear before a post-cutoff observation. If the reform itself affected continued reporting or survival differently across treatment groups, this rule could influence the estimates. Complete-case requirements also produce somewhat different leverage, risk, and cash samples. This is especially relevant for $EarnVol$, whose forward three-year window requires later accounting observations. Cross-outcome differences therefore combine behavioral differences with some variation in the firms represented, whereas inference within each outcome remains tied to its reported estimation sample.

8.2 Treatment Measurement and Identification

The country classification provides a transparent way to operationalize the legal change, but it measures exposure to a new policy environment rather than whether the statutory

cap bound a particular firm. Under the architecture documented in Section 2.4, actual binding depends on statutory measures of exceeding borrowing costs and tax-adjusted EBITDA, unused interest capacity, and applicable national options that are not observed in the accounting data. The main coefficient should therefore be read as the average response to assignment to the newly exposed group, not as the effect on firms known to have lost an interest deduction. If many treated firms were not constrained, the average response could be smaller than the response among actually affected firms; heterogeneous implementations may also contribute to differences across countries. The regime and accounting-pressure analyses help assess this variation, but the accounting measure remains only a proxy for firm-level exposure.

Because the analysis uses consolidated accounts, it measures financing changes at the corporate-group level. This prevents a mere relocation of debt between entities under common ownership from appearing as a reduction in the group's overall leverage. The remaining limitation concerns the measurement of policy exposure rather than the accounting unit. Treatment is assigned by headquarters country, although multinational groups can operate and borrow through subsidiaries in several jurisdictions. The coefficient therefore captures changes in consolidated outcomes associated with headquarters-country exposure, but it cannot show where within the group the adjustment occurred or what share of the group was directly exposed to the new rule. This does not invalidate the consolidated estimate; it narrows its interpretation and prevents a subsidiary-level or jurisdiction-specific conclusion.

As in any difference-in-differences design based on country-level reforms, causal interpretation depends on comparable counterfactual trends. Firm fixed effects remove stable firm characteristics, year fixed effects absorb common shocks, and the country controls, matching, and ATT reweighting address several observable differences. These steps materially improve comparability, but they cannot account for every unobserved country-specific development or simultaneous reform. The leverage diagnostics provide qualified support for the design: the joint pre-trend test does not reject, and the negative coefficient remains across the main robustness checks. At the same time, the marginally significant 2015 and 2016 coefficients indicate some movement before implementation, while equity and especially long-term debt display weaker pre-period dynamics (Appendix Table 18). The leverage result is therefore best viewed as a robust relative pattern that is consistent with the proposed causal channel, rather than as evidence that excludes every competing country-level explanation. The component estimates reinforce the balance-sheet interpretation, but their weaker dynamics prevent them from providing separate causal confirmation.

A separate timing limitation applies to the operational-risk analysis. Whereas leverage and cash use the 2019 implementation benchmark, operational risk uses 2017 as a common analytical cutoff following the reference design rather than a country-specific event date. Each $EarnVol_t$ observation summarizes performance from t through $t + 2$, so adjacent

observations share two accounting years. Consequently, the outcomes assigned to the 2015 and 2016 base years already contain performance from 2017, and the 2016 outcome also contains 2018. A change in a single accounting year can therefore affect several adjacent outcome windows, blurring the timing of any response and making the pre-cutoff coefficients less informative as conventional pre-trend diagnostics. Together with the borderline adjusted-risk pre-trend and the rejecting matched and weighted tests, this feature is an important reason to interpret the operational-risk result as suggestive rather than decisive.

Beyond this $EarnVol$-specific issue, the post-reform outcomes also overlap the COVID-19 pandemic. The largest leverage decline occurs in 2020, and excluding 2020–2021 attenuates the estimate from -2.05 to -1.50 percentage points. The negative leverage estimate is therefore not confined to those two years, although its smaller magnitude in the restricted sample reduces confidence in the exact size of the response. Specifications that retain the full samples and control jointly for national COVID-19 mortality and government stringency produce leverage and operational-risk estimates close to those from the corresponding main specifications. These controls do not explain the estimated declines, but their national and aggregated nature cannot capture firm- or industry-specific pandemic exposure. The pandemic overlap therefore remains a qualification for the magnitude of the leverage result and for a strong interpretation of the operational-risk result.

8.3 Statistical Precision

The regressions contain many firm-year observations, but treatment varies at the country level. The relevant independent policy variation therefore comes from the 20 headquarters-country clusters in the main specifications. Clustering at the country level is the appropriate choice, but 20 clusters provide a relatively limited basis for conventional cluster-robust inference. The reported standard errors and significance levels should therefore be interpreted cautiously, especially for estimates near conventional significance thresholds. This concern is more pronounced in subgroup robustness checks with still fewer country clusters, whose estimates are accordingly treated as suggestive. This limitation concerns statistical precision; it does not by itself alter the point estimates or demonstrate coefficient bias.

9 Discussion and Policy Implications

The empirical analysis first asks whether newly exposed firms exhibit the financing response predicted by an earnings-stripping rule and then examines whether adjustment extends to operating stability and liquidity. Section 2.4 documents the 30-percent headline threshold and the national options surrounding it. The design does not observe whether the statutory cap binds a particular firm. Instead, it compares changes among firms headquartered in countries without a pre-2019 regime classified as equally effective with changes among firms in countries where such a regime already existed. Within this comparison, the leverage analysis examines the financing response most closely linked to tighter deductibility; the earnings-volatility and cash analyses then assess possible consequences beyond that margin.

9.1 Interpretation Across Outcomes

Among the three outcomes, capital structure provides the most consistent evidence for the financing response predicted by the framework. Newly exposed firms reduce debt-to-assets by 2.05 percentage points relative to firms headquartered in countries with a pre-existing regime classified as equally effective. Together with higher equity and lower current and long-term debt, this pattern supports H1 and is consistent with a broader movement away from debt financing. In this narrow sense, the result is consistent with the intended financing incentive of tighter deductibility; it is not a comprehensive test of the rule's success in raising revenue or reducing base erosion. The concentration of the largest dynamic estimate in 2020, weaker pre-reform dynamics for some component outcomes, and the limited number of country-level policy clusters continue to prevent an unqualified causal interpretation.

The negative adjusted-$EarnVol$ estimate is consistent with more stable subsequent operating performance when deduction capacity becomes more sensitive to earnings. This matches the state-dependent mechanism because lower volatility can reduce the probability that a given interest burden becomes constrained in a low-profit state. The outcome, however, measures realized accounting volatility rather than managerial choices. Its marginal precision, differential pre-cutoff dynamics, and attenuation in the strict pre-pandemic check mean that H2 remains suggestive rather than evidence of deliberate risk reduction.

The cash results do not support H3's predicted positive average accumulation. This neither establishes a zero effect nor makes liquidity irrelevant. Cash/assets is a net stock, and the main-specification cash/assets estimate is conditional on contemporaneous leverage. Debt repayment is therefore only one possible, unidentified explanation for the absence of accumulation. The evidence consequently cannot establish whether leverage, operating stability, and liquidity complement or substitute for one another within the same firms.

9.2 Comparison with Prior Evidence

The leverage finding fits the distinction between narrow thin-capitalization rules and broader restrictions on total net interest. Related-party rules can reduce internal debt while leaving scope for external-debt substitution (Buettner et al., 2012; Wamser, 2014), whereas broader rules are more likely to affect consolidated leverage (Mooij and Hebous, 2018). The debt-to-assets estimate from the main specification is close in magnitude to the ATAD-era response reported by De Simone et al. (2025): a 2.05-percentage-point decline in this thesis compared with 2.3 percentage points in that study. Differences in data and specification mean that this similarity should be treated as a benchmark rather than evidence of equivalent effects. The result contrasts with evidence from Finland's earlier earnings-based interest barrier, which reduced financial expenses but produced no significant baseline response in total debt (Harju et al., 2025). This contrast indicates that broad interest limitations do not uniformly induce deleveraging and that the financing margin observed in this thesis may depend on rule design and firms' effective exposure.

The risk result likewise belongs to a mixed literature on real responses. Its negative adjusted-*EarnVol* estimate is directionally consistent with De Simone et al. (2025) (-1.96 percentage points compared with -0.657 percentage points), but differences in measurement and the present analysis's weaker dynamic diagnostics make the gap in absolute magnitude unsuitable as a direct comparison of response strength. Evidence on other real outcomes is also mixed: Leszczyłowska and Meier (2021) estimate declines in net investment, whereas Harju et al. (2025) find no detectable response in EBITDA or turnover. The cash analysis adds a distinct margin without a directly comparable coefficient in these studies and yields no affirmative average response. Considering all three outcomes together shows why financing, operating stability, and liquidity must be assessed separately rather than inferred from leverage alone.

Direct evidence on interest limitations sharpens rather than resolves this ambiguity. Broad restrictions covering total debt have been associated with lower leverage and predicted financial distress, supporting a resilience channel, while European earnings-stripping rules and the U.S. interest cap have also been linked to lower investment (Mooij and Hebous, 2018; Leszczyłowska and Meier, 2021; Sanati and Beyhaghi, 2025). Conversely, Finnish and U.S. evidence documents small or statistically insignificant average real responses in other settings (Harju et al., 2025; Goodman et al., 2025). These differences suggest that binding exposure, rule design, and firms' financing alternatives condition the trade-off.

In this thesis, the leverage decline, tentative volatility response, and absence of affirmative average cash accumulation provide diagnostic evidence across separate adjustment margins; they do not reveal whether firms became more resilient or abandoned valuable risky activity. Policy evaluation should therefore combine financing and volatility outcomes with direct measures of investment, R&D, patents, growth, distress, and tax revenue.

9.3 Policy Implications

The leverage evidence offers a qualified policy implication for financing. The consolidated leverage pattern is consistent with broad earnings-based limits affecting total group financing rather than only debt placement between related entities. In the pre-reform-policy-regime check reported in Table 9, the largest leverage point estimate occurs among countries without a prior interest-limitation rule. This pattern is consistent with stronger adjustment where the change in legal exposure was greatest. Because consolidated accounts do not reveal whether MNEs reallocated debt among affiliates, policy evaluations should examine intra-group financing flows alongside consolidated debt and equity.

Statutory design and effective exposure nevertheless matter. The national options documented in Section 2.4, together with pre-existing regimes, can separate formal adoption from firm-level binding. The headquarters-country estimate cannot determine an optimal threshold or the effects of individual relief provisions. The suggestive volatility result indicates that a rule directed at interest deductibility may also be associated with operating adjustment, but it does not establish safer investment, improved financial stability, or welfare gains.

The suggestive decline in adjusted earnings volatility raises a stability–dynamism trade-off, but lower realized volatility is not itself a welfare criterion. Deleveraging can reduce firms' sensitivity to adverse shocks, while productive risk management can stabilize internal funds and preserve valuable investment when external finance is costly (Giroud and Mueller, 2017; Froot et al., 1993; Pérez-González and Yun, 2013). Conversely, stronger downside penalties can reduce cash-flow risk by encouraging conservative or even value-reducing project choices, and innovation often requires tolerance of early failure (Acharya et al., 2011; Manso, 2011). The same accounting pattern could therefore reflect greater resilience, less experimentation, or a combination of both.

These trade-offs define a more targeted research agenda. Future work should combine longer post-reform panels with direct measures of investment, R&D, patents, growth, and financial distress. Additional accounting years would show whether the leverage and earnings-volatility patterns persist beyond the initial adjustment and pandemic periods, while providing more complete volatility windows for a cleaner COVID-period sensitivity analysis. Linking these outcomes would help distinguish improved resilience from reduced experimentation rather than inferring the mechanism from lower accounting volatility alone.

10 Conclusion

Modern earnings-stripping rules make the tax value of debt contingent on operating earnings. The institutional reform examined here is the national implementation of Article 4 of Council Directive (EU) 2016/1164, the Anti-Tax Avoidance Directive (ATAD). Under its headline cap, exceeding borrowing costs are deductible only up to 30 percent of tax-adjusted EBITDA, subject to national implementation options and relief provisions. Because deduction capacity consequently varies with operating earnings, the debt tax shield becomes less reliable in low-profit states and firms may adjust financing, operating stability, or liquidity. This thesis examines whether firms newly exposed to the ATAD-era regime exhibit the predicted financing response and whether adjustment extends beyond leverage to subsequent earnings volatility and cash holdings. Using consolidated accounting data for European non-financial firms, the empirical design compares within-firm changes in newly exposed countries with those in countries that already had an interest-limitation regime classified as equally effective. The estimates are therefore conditional relative changes, not universal effects or direct observations of managerial intent.

Capital structure provides the clearest and most directionally stable evidence. After the 2019 implementation benchmark, newly exposed firms reduce debt-to-assets by an estimated 2.05 percentage points relative to the control group, equivalent to approximately 9% of the descriptive sample mean. Equity increases while current and long-term debt decrease, a pattern consistent with a broader movement away from debt financing rather than only a shift in debt maturity. The component estimates do not, however, identify the financing flows that produced this adjustment. The debt-to-assets estimate remains negative across the selected robustness checks, although the pandemic years contribute to its magnitude and the design cannot exclude every competing country-level explanation. It is therefore best understood as a robust relative pattern consistent with the proposed financing channel rather than as unqualified causal proof.

Evidence of adjustment beyond financing is more limited. For the primary adjusted earnings-volatility measure, *EarnVol*, the estimated effect is -1.96 percentage points, which has the predicted sign and is economically non-trivial but imprecisely estimated. It is consistent with more stable subsequent operating performance relative to the country–two-digit-industry–year peer benchmark. However, the differential pre-cutoff dynamics, changing precision across specifications, and attenuation in the strict pre-pandemic check prevent interpreting the result as evidence of deliberate managerial risk reduction or a decisive causal mechanism. The cash/assets estimate from the main specification is small, negative, and imprecise at -0.39 percentage points, and none of the selected robustness checks reveals a positive and statistically precise average response. The analysis therefore does not support the predicted accumulation of precautionary cash, but it neither establishes a zero effect nor shows that the reform caused cash holdings to fall.

Considering the three outcomes jointly shows why financing, operating stability, and liquidity must be assessed separately rather than inferred from leverage alone. The state-

dependent mechanism allows the margins to complement or substitute for one another, but the evidence cannot establish whether they did so within the same firms. The financing pattern fits the distinction between broad restrictions on total net interest and narrower related-party rules: the former can affect consolidated financing, whereas the latter may primarily alter debt composition. Accordingly, policy evaluation should examine consolidated debt and equity alongside related-party interest flows.

Several limitations govern the strength and scope of these conclusions. Treatment captures headquarters-country exposure to a new policy environment, not whether the statutory cap bound a particular firm; statutory measures of exceeding borrowing costs and tax-adjusted EBITDA, unused interest capacity, and applicable national options are not directly observed. Consolidated accounts reveal group-level outcomes but not where within a multinational group adjustment occurred. Treatment also varies across only 20 headquarters-country clusters. The fixed effects, controls, diagnostics, and sensitivity analyses strengthen the comparison but cannot remove every simultaneous reform or unobserved country-specific development. Pandemic overlap and the forward-looking risk measure create additional timing concerns. The estimates therefore apply most directly to the represented European non-financial firms with consolidated accounts rather than to all European firms or to firms outside Europe.

The policy implications are correspondingly narrow. The consolidated financing pattern is consistent with broad earnings-based limits affecting total group financing, while the suggestive decline in adjusted earnings volatility indicates that adjustment may extend beyond capital structure. Lower realized volatility, however, is not itself a welfare criterion: it could reflect improved resilience, reduced experimentation, or both. The analysis does not establish effects on tax revenue, base erosion, investment, innovation, or optimal thresholds, nor does it provide a basis for presuming that firms build additional observable cash buffers. Future work should combine longer post-reform panels with direct measures of investment, R&D, growth, and financial distress. Additional accounting years would help determine whether the leverage and earnings-volatility patterns persist beyond the initial adjustment and pandemic periods and would provide more complete volatility windows for a cleaner COVID-period sensitivity analysis. Taken together, the evidence is strongest for a relative shift away from debt financing; beyond that margin, the thesis identifies a tentative decline in adjusted earnings volatility but no affirmative evidence of positive average cash accumulation.

Appendix

Data and Sample

These tables supplement the data description in Section 4.

Table 11: Earnings-stripping-rule design in the baseline countries

Country	Group	EBITDA rule	De minimis	Group escape	Interest carryforward	EBITDA carryforward
Belgium (BE)	Treatment	Yes	3m	X	Indefinite	X
Bulgaria (BG)	Treatment	Yes	3m	X	Indefinite	X
Cyprus (CY)	Treatment	Yes	3m	Yes	5 years	5 years
Czech Republic (CZ)	Treatment	Yes	3.2m	X	Indefinite	X
Denmark (DK)	Treatment	Yes	3m	X	Indefinite	5 years
Estonia (EE)	Treatment	Yes	3m	Yes	X	X
Finland (FI)	Treatment	25%/EBITDA	3m	Yes	Indefinite	X
Croatia (HR)	Treatment	Yes	3m	X	3 years	X
Hungary (HU)	Treatment	Yes	2.5m	Yes	Indefinite	Indefinite
Italy (IT)	Treatment	Yes	X	X	Indefinite	5 years
Lithuania (LT)	Treatment	Yes	3m	Yes	Indefinite	X
Luxembourg (LU)	Treatment	Yes	3m	Yes	Indefinite	5 years
Malta (MT)	Treatment	Yes	3m	Yes	Indefinite	5 years
Netherlands (NL)	Treatment	Yes	1m	X	Indefinite	X
Portugal (PT)	Treatment	Yes	1m	X	5 years	5 years
Sweden (SE)	Treatment	Yes	0.45m	X	6 years	X
Germany (DE)	Control	Yes	3m	Yes	Indefinite	5 years
Spain (ES)	Control	Yes	1m	X	Indefinite	5 years
France (FR)	Control	Yes	3m	Yes	Indefinite	5 years
Greece (GR)	Control	Yes	3m	X	Indefinite	X
Slovenia (SI)	Control	Yes	1m	X	X	X
Slovak Republic (SK)	Control	Yes	3m	X	5 years	X

Notes: Source: Adapted from Tables A1 and A2 in De Simone et al. (2025). Group assignments follow the pre-reform classification in their Table A1: control countries had an interest-limitation rule in line with the EU Directive, whereas treatment countries did not. The remaining columns report national earnings-stripping-rule features at implementation from their Table A2. X denotes that no such provision is recorded; Indefinite denotes an indefinite carryforward period.

Table 12: Pre-reform treated-control differences in firm-level means

Role	Variable	Treated firms	Control firms	Treated mean	Control mean	Difference
Capital structure	Debt-to-assets	1,672	1,120	0.214	0.240	-0.026
Liquidity	Cash/assets	1,669	1,119	0.184	0.161	0.023
Risk	Earnings volatility	1,653	1,112	14.062	7.535	6.528
Growth	Log sales growth	1,489	1,048	0.103	0.053	0.051
Firm controls	EBIT/assets	1,658	1,114	-0.054	0.010	-0.064
Firm controls	Tangibility	1,667	1,118	0.183	0.207	-0.024
Firm controls	Log assets	1,673	1,121	4.262	5.299	-1.037

Notes: The pre-reform period covers firm-year observations through 2018.

Table 13: Country-level sample composition

Country	Group	Observations	Firms	First year	Last year
DE	Control	3,356	349	2014	2023
ES	Control	1,524	165	2014	2023
FR	Control	4,384	458	2014	2023
GR	Control	1,123	116	2014	2023
SI	Control	115	13	2014	2023
SK	Control	196	20	2014	2023
BE	Treatment	962	100	2014	2023
BG	Treatment	724	76	2014	2023
CY	Treatment	583	63	2014	2023
CZ	Treatment	107	13	2014	2023
DK	Treatment	1,033	113	2014	2023
EE	Treatment	165	17	2014	2023
FI	Treatment	1,199	129	2014	2023
HR	Treatment	442	45	2014	2023
HU	Treatment	236	26	2014	2023
IT	Treatment	1,892	206	2014	2023
LT	Treatment	218	23	2014	2023
LU	Treatment	368	39	2014	2023
MT	Treatment	236	26	2014	2023
NL	Treatment	986	105	2014	2023
PT	Treatment	268	27	2014	2023
SE	Treatment	6,095	665	2014	2023

Notes: Country refers to the firm headquarters country in the 2014–2023 baseline firm-year panel.

Baseline Results

These tables and the figure supplement the baseline results in Section 6.

Table 14: Capital-structure event-study regressions

Model:	Debt/assets (1)	Log equity (2)	Log current debt (3)	Log long-term debt (4)
Variables				
Treat × 2014	0.0093	-0.0027	0.0105	-0.0098
	(0.0115)	(0.0338)	(0.1178)	(0.0726)
Treat × 2015	0.0131*	-0.0431	-0.0418	0.1358**
	(0.0074)	(0.0264)	(0.1045)	(0.0566)
Treat × 2016	0.0141*	-0.0315	-0.0833	0.1365**
	(0.0070)	(0.0212)	(0.0859)	(0.0583)
Treat × 2017	0.0042	-0.0129	-0.0567	-0.0036
	(0.0045)	(0.0192)	(0.0711)	(0.0526)
Treat × 2019	-0.0063	0.0169	0.0046	-0.0205
	(0.0065)	(0.0160)	(0.0654)	(0.0737)
Treat × 2020	-0.0276***	0.0775***	-0.1153	-0.1924**
	(0.0093)	(0.0195)	(0.0827)	(0.0815)
Treat × 2021	-0.0185*	0.0387	-0.2240**	-0.1446
	(0.0100)	(0.0268)	(0.0950)	(0.0852)
Treat × 2022	-0.0060	0.0260	-0.1824	-0.0607
	(0.0108)	(0.0163)	(0.1260)	(0.0822)
Treat × 2023	-0.0058	0.0102	-0.2381**	-0.1032
	(0.0115)	(0.0165)	(0.1061)	(0.0979)
Tangibility	0.2111***	-0.5680***	0.3512**	1.693***
	(0.0325)	(0.1185)	(0.1595)	(0.4100)
EBITDA/assets	-0.1094***	0.3385***	-0.8930***	-0.9136***
	(0.0276)	(0.0778)	(0.2104)	(0.1057)
Log assets	0.0120	1.021***	0.8858***	1.016***
	(0.0076)	(0.0302)	(0.0264)	(0.0609)
Country-sector sales growth	-0.0009	-0.0438***	-0.0042	0.0897**
	(0.0031)	(0.0143)	(0.0401)	(0.0363)
Corporate tax rate	-0.0015**	0.0016	-0.0026	-0.0103***
	(0.0006)	(0.0012)	(0.0065)	(0.0032)
GDP growth	-0.0003	0.0004	-0.0123*	0.0008
	(0.0006)	(0.0019)	(0.0062)	(0.0038)
Inflation	0.0010	-0.0088**	0.0068	0.0062
	(0.0010)	(0.0032)	(0.0165)	(0.0120)
Long-term interest rate	0.0022	0.0090**	0.0517**	-0.0084
	(0.0016)	(0.0037)	(0.0187)	(0.0156)
Rule of law	-0.0027	0.0009	-0.0069	-0.0076
	(0.0021)	(0.0043)	(0.0164)	(0.0178)
Regulatory quality	0.0015	-0.0020	0.0271**	0.0207**
	(0.0013)	(0.0029)	(0.0109)	(0.0086)
Control of corruption	-7.87×10^{-5}	-0.0040*	-0.0037	0.0065
	(0.0007)	(0.0021)	(0.0106)	(0.0063)
Fixed-effects				
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Fit statistics				
Firms	2,655	2,591	2,150	2,215
Years	10	10	10	10
Observations	24,851	23,745	18,634	19,622
Adjusted R^2	0.681	0.979	0.832	0.907
Within R^2	0.042	0.653	0.100	0.198

Notes: The table reports the main full-control event-study specifications. The coefficients are treated-by-year interactions relative to the omitted reference year 2018. Standard errors are clustered by headquarters country (20 clusters) and appear in parentheses. The debt/assets coefficients are reported in ratio units; multiplying them by 100 gives effects in percentage points (pp). Coefficients for the other outcomes are in log points. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 15: Earnings-volatility event-study regressions

Model:	Adjusted EarnVol (1)	Unadjusted EarnVol (2)
Variables		
Treat × 2013	-0.7385	0.1089
	(1.342)	(0.4791)
Treat × 2014	-1.091	0.1203
	(1.171)	(0.4788)
Treat × 2015	0.4758	0.6482*
	(0.8390)	(0.3277)
Treat × 2017	-1.027	-0.5831*
	(0.8203)	(0.3133)
Treat × 2018	-2.809***	-0.8950***
	(0.8698)	(0.1969)
Treat × 2019	-3.786***	-1.084***
	(0.8832)	(0.2776)
Treat × 2020	-2.888***	-0.4086
	(0.7518)	(0.4209)
Treat × 2021	-1.042	-0.1333
	(0.6303)	(0.3022)
Log assets	-3.900***	-2.742***
	(0.6153)	(0.2847)
EBIT/assets	-19.08***	-19.41***
	(2.648)	(1.391)
Log sales growth	-0.7559***	-0.2338
	(0.2551)	(0.1459)
Log sales	3.230***	2.244***
	(0.8065)	(0.3569)
Corporate tax rate	-0.0586	-0.0361
	(0.1086)	(0.0416)
GDP growth	-0.0221	-0.0517
	(0.0825)	(0.0365)
Inflation	0.3144	-0.0179
	(0.4034)	(0.1728)
Long-term interest rate	-0.0048	-0.1356*
	(0.1647)	(0.0705)
Rule of law	-0.0705	0.0055
	(0.1671)	(0.0763)
Regulatory quality	0.0032	0.0028
	(0.1146)	(0.0721)
Control of corruption	0.0645	0.0268
	(0.1045)	(0.0489)
Fixed-effects		
Firm	Yes	Yes
Year	Yes	Yes
Fit statistics		
Firms	2,183	2,183
Years	9	9
Observations	18,562	18,562
Adjusted R^2	0.492	0.593
Within R^2	0.043	0.102

Notes: The table reports the main full-control event-study specifications. The coefficients are treated-by-base-year interactions relative to the omitted reference base year 2016. Each base year marks the beginning of a forward three-year window covering years t through $t + 2$. Standard errors are clustered by headquarters country (20 clusters) and appear in parentheses. Both outcomes equal 100 times the forward-looking three-year standard deviation of EBIT/assets, so coefficients are expressed in percentage points (pp). Adjusted EarnVol removes the country-industry-year benchmark before volatility is calculated. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 16: Cash-holdings regression

Model:	Cash/assets (1)
Variables	
Treat × Post2019	-0.0039
	(0.0068)
Log assets	-0.0105***
	(0.0035)
EBIT/assets	0.0439***
	(0.0104)
Tangibility	-0.2294***
	(0.0233)
Debt-to-assets	-0.1193***
	(0.0220)
Log sales growth	-0.0049***
	(0.0012)
Corporate tax rate	-0.0015***
	(0.0004)
GDP growth	0.0016***
	(0.0004)
Inflation	0.0004
	(0.0018)
Long-term interest rate	0.0032
	(0.0020)
Rule of law	0.0013
	(0.0013)
Regulatory quality	0.0004
	(0.0008)
Control of corruption	8.9×10^{-6}
	(0.0007)
Fixed-effects	
Firm	Yes
Year	Yes
Fit statistics	
Firms	2,427
Years	10
Observations	22,570
Adjusted R^2	0.729
Within R^2	0.069

Notes: The table reports the main full-control specification with firm and year fixed effects. Standard errors are clustered by headquarters country (20 clusters). $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 17: Cash-holdings event-study regression

Model:	Cash/assets (1)
Variables	
Treat × 2014	-0.0056
	(0.0060)
Treat × 2015	-0.0070
	(0.0066)
Treat × 2016	0.0004
	(0.0068)
Treat × 2017	-0.0087
	(0.0062)
Treat × 2019	-0.0029
	(0.0043)
Treat × 2020	-0.0098
	(0.0110)
Treat × 2021	-0.0050
	(0.0086)
Treat × 2022	-0.0080
	(0.0080)
Treat × 2023	-0.0129
	(0.0106)
Log assets	-0.0104***
	(0.0036)
EBIT/assets	0.0437***
	(0.0106)
Tangibility	-0.2293***
	(0.0233)
Debt-to-assets	-0.1194***
	(0.0220)
Log sales growth	-0.0049***
	(0.0012)
Corporate tax rate	-0.0016***
	(0.0004)
GDP growth	0.0015***
	(0.0005)
Inflation	0.0007
	(0.0017)
Long-term interest rate	0.0033
	(0.0019)
Rule of law	0.0011
	(0.0011)
Regulatory quality	0.0005
	(0.0008)
Control of corruption	0.0003
	(0.0008)
Fixed-effects	
Firm	Yes
Year	Yes
Fit statistics	
Firms	2,427
Years	10
Observations	22,570
Adjusted R^2	0.729
Within R^2	0.070

Notes: The table reports the main full-control event-study specification. The coefficients are treated-by-year interactions relative to the omitted reference year 2018. Standard errors are clustered by headquarters country (20 clusters) and appear in parentheses. Cash/assets coefficients are reported in ratio units; multiplying them by 100 gives effects in percentage points (pp). $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

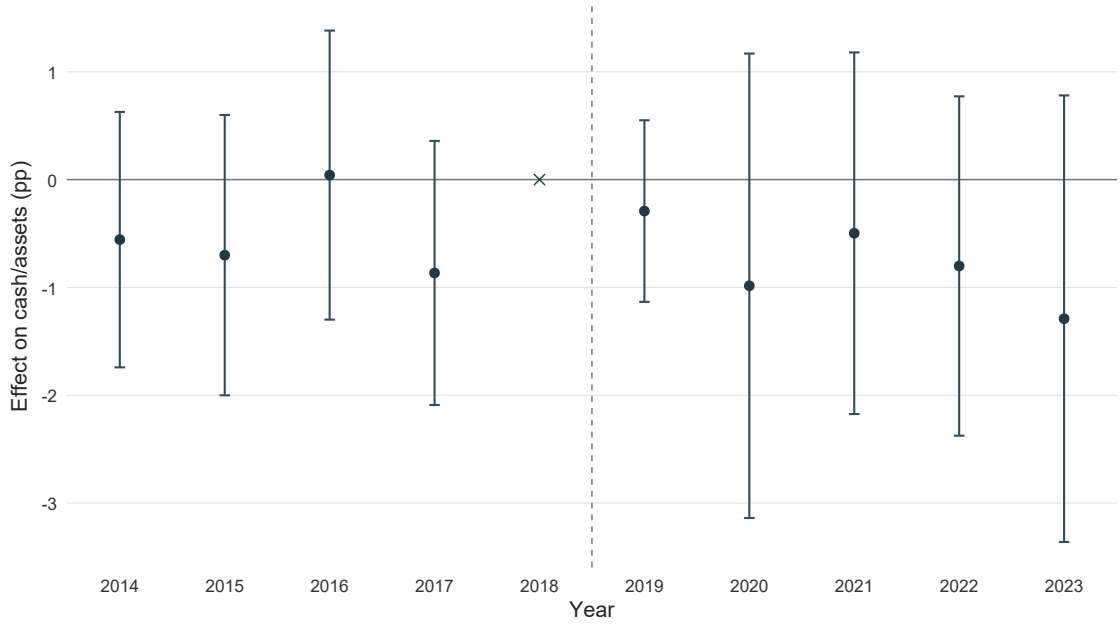


Figure 3: Event-study estimates for cash/assets

Table 18: Joint pre-trend tests for baseline event studies

Outcome	Omitted year	Tested years	Wald χ^2	df	p-value
Capital structure					
Debt-to-assets	2018	2014–2017	6.043	4	0.1959
Log equity	2018	2014–2017	8.436	4	0.0768
Log current debt	2018	2014–2017	3.599	4	0.4630
Log long-term debt	2018	2014–2017	36.475	4	< 0.0001
Operational risk					
Adjusted *EarnVol*	2016	2013–2015	6.658	3	0.0836
Unadjusted *EarnVol*	2016	2013–2015	13.345	3	0.0039
Cash holdings					
Cash/assets	2018	2014–2017	4.210	4	0.3783

Notes: The table reports joint tests of whether the indicated pre-cutoff event-study coefficients equal zero relative to the omitted year. The models follow the corresponding baseline specifications, with inference clustered by headquarters country. For the operational-risk outcomes, the tested years mark the beginning of forward-looking three-year windows. A statistically insignificant result does not establish parallel trends.

Robustness

These tables provide the detailed results discussed in Section 7.

Control-Set Sensitivity

Table 19: Debt-to-assets regressions across control sets

Model:	Firm controls (1)	No interest rate (2)	Full controls (3)
Variables			
Treat × Post2019	-0.0278**	-0.0193**	-0.0205***
	(0.0113)	(0.0070)	(0.0068)
Tangibility	0.2089***	0.2141***	0.2114***
	(0.0319)	(0.0323)	(0.0326)
EBITDA/assets	-0.1104***	-0.1095***	-0.1096***
	(0.0271)	(0.0277)	(0.0276)
Log assets	0.0126	0.0114	0.0118
	(0.0075)	(0.0076)	(0.0077)
Country-sector sales growth	0.0005	-0.0003	-0.0008
	(0.0029)	(0.0030)	(0.0030)
Corporate tax rate		-0.0013**	-0.0013**
		(0.0006)	(0.0005)
GDP growth		-0.0010	-0.0009
		(0.0008)	(0.0008)
Inflation		0.0017*	0.0017*
		(0.0009)	(0.0010)
Rule of law		-0.0033	-0.0031
		(0.0023)	(0.0021)
Regulatory quality		0.0017	0.0019*
		(0.0011)	(0.0011)
Control of corruption		0.0002	0.0002
		(0.0007)	(0.0008)
Long-term interest rate			0.0017
			(0.0015)
Fixed-effects			
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Firms	2,728	2,672	2,655
Years	10	10	10
Observations	25,527	25,011	24,851
Adjusted R^2	0.682	0.681	0.681
Within R^2	0.038	0.041	0.041

Notes: The table reports complete coefficient estimates for the selected robustness specifications. Firm and year fixed effects are included throughout. Standard errors are clustered by headquarters country (20–22 clusters across the reported models) and appear in parentheses. The dependent variable is debt-to-assets; coefficients are in ratio units. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 20: Adjusted earnings-volatility regressions across control sets

Model:	Firm controls (1)	Macro/tax controls (2)	Full controls (3)
Variables			
Treat × Reform2017	-1.672	-2.059**	-1.956*
	(0.9978)	(0.9728)	(1.027)
Log assets	-3.854***	-3.867***	-3.866***
	(0.5782)	(0.6218)	(0.6234)
EBIT/assets	-19.17***	-19.08***	-19.07***
	(2.679)	(2.662)	(2.656)
Log sales growth	-0.5705*	-0.7457***	-0.7505***
	(0.2752)	(0.2554)	(0.2554)
Log sales	2.857***	3.223***	3.226***
	(0.8065)	(0.8077)	(0.8083)
Corporate tax rate		-0.0805	-0.0813
		(0.0831)	(0.0940)
GDP growth		-0.0471	-0.0569
		(0.0557)	(0.0536)
Inflation		0.2723	0.2949
		(0.4378)	(0.4445)
Long-term interest rate		-0.0521	-0.0345
		(0.1298)	(0.1414)
Rule of law			-0.0373
			(0.1770)
Regulatory quality			0.0211
			(0.1178)
Control of corruption			0.0799
			(0.1157)
Fixed-effects			
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Firms	2,244	2,183	2,183
Years	9	9	9
Observations	19,083	18,562	18,562
Adjusted R^2	0.484	0.492	0.492
Within R^2	0.037	0.042	0.042

Notes: The table reports complete coefficient estimates for the selected robustness specifications. Firm and year fixed effects are included throughout. Standard errors are clustered by headquarters country (20–22 clusters across the reported models) and appear in parentheses. The dependent variable is adjusted earnings volatility and is expressed in percentage points. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 21: Cash/assets regressions across control sets

Model:	Firm controls (1)	Macro/tax controls (2)	Full controls (3)
Variables			
Treat × Post2019	-0.0053	-0.0030	-0.0039
	(0.0096)	(0.0071)	(0.0068)
Log assets	-0.0111***	-0.0108***	-0.0105***
	(0.0036)	(0.0036)	(0.0035)
EBIT/assets	0.0444***	0.0440***	0.0439***
	(0.0103)	(0.0104)	(0.0104)
Tangibility	-0.2320***	-0.2306***	-0.2294***
	(0.0234)	(0.0234)	(0.0233)
Debt-to-assets	-0.1178***	-0.1196***	-0.1193***
	(0.0219)	(0.0221)	(0.0220)
Log sales growth	-0.0046***	-0.0048***	-0.0049***
	(0.0012)	(0.0012)	(0.0012)
Corporate tax rate		-0.0015***	-0.0015***
		(0.0004)	(0.0004)
GDP growth		0.0018***	0.0016***
		(0.0005)	(0.0004)
Inflation		0.0006	0.0004
		(0.0017)	(0.0018)
Long-term interest rate		0.0023	0.0032
		(0.0017)	(0.0020)
Rule of law			0.0013
			(0.0013)
Regulatory quality			0.0004
			(0.0008)
Control of corruption			8.9×10^{-6}
			(0.0007)
Fixed-effects			
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Firms	2,491	2,427	2,427
Years	10	10	10
Observations	23,172	22,570	22,570
Adjusted R^2	0.730	0.729	0.729
Within R^2	0.067	0.069	0.069

Notes: The table reports complete coefficient estimates for the selected robustness specifications. Firm and year fixed effects are included throughout. Standard errors are clustered by headquarters country (20–22 clusters across the reported models) and appear in parentheses. The dependent variable is cash/assets; coefficients are in ratio units. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Matching and ATT Reweighting

Table 22: Debt-to-assets regressions after matching and reweighting

Model:	Main model (1)	PS match (2)	ATT weighted (3)
Variables			
Treat × Post2019	-0.0205***	-0.0143**	-0.0220***
	(0.0068)	(0.0066)	(0.0060)
Tangibility	0.2114***	0.1991***	0.2097***
	(0.0326)	(0.0407)	(0.0384)
EBITDA/assets	-0.1096***	-0.1265***	-0.1033***
	(0.0276)	(0.0279)	(0.0222)
Log assets	0.0118	0.0167	0.0047
	(0.0077)	(0.0113)	(0.0127)
Country-sector sales growth	-0.0008	-0.0031	-0.0035
	(0.0030)	(0.0032)	(0.0035)
Corporate tax rate	-0.0013**	-0.0019***	-0.0014***
	(0.0005)	(0.0004)	(0.0005)
GDP growth	-0.0009	-0.0013	-0.0011
	(0.0008)	(0.0009)	(0.0007)
Inflation	0.0017*	0.0025**	0.0018*
	(0.0010)	(0.0012)	(0.0009)
Long-term interest rate	0.0017	0.0019	0.0022
	(0.0015)	(0.0015)	(0.0013)
Rule of law	-0.0031	-0.0042*	-0.0057**
	(0.0021)	(0.0022)	(0.0023)
Regulatory quality	0.0019*	0.0008	0.0027**
	(0.0011)	(0.0012)	(0.0012)
Control of corruption	0.0002	0.0012*	0.0007
	(0.0008)	(0.0007)	(0.0008)
Fixed-effects			
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Firms	2,655	1,861	2,646
Years	10	10	10
Observations	24,851	17,588	24,793
Adjusted R^2	0.681	0.700	0.672
Within R^2	0.041	0.042	0.040

Notes: The table reports complete coefficient estimates for the selected robustness specifications. Firm and year fixed effects are included throughout. Standard errors are clustered by headquarters country (20 clusters across the reported models) and appear in parentheses. The dependent variable is debt-to-assets; coefficients are in ratio units. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 23: Adjusted earnings-volatility regressions after matching and reweighting

Model:	Main model (1)	PS match (2)	ATT weighted (3)
Variables			
Treat × Reform2017	-1.956*	-1.854*	-1.397
	(1.027)	(1.062)	(0.8851)
Log assets	-3.866***	-4.104***	-3.706***
	(0.6234)	(0.5526)	(0.5199)
EBIT/assets	-19.07***	-15.05***	-18.70***
	(2.656)	(2.657)	(2.126)
Log sales growth	-0.7505***	-0.6751*	-0.6864**
	(0.2554)	(0.3741)	(0.2528)
Log sales	3.226***	2.908***	2.785***
	(0.8083)	(0.8809)	(0.5756)
Corporate tax rate	-0.0813	-0.0462	-0.0330
	(0.0940)	(0.0753)	(0.0733)
GDP growth	-0.0569	-0.0448	-0.0503
	(0.0536)	(0.0474)	(0.0499)
Inflation	0.2949	0.4164	0.1637
	(0.4445)	(0.4075)	(0.4396)
Long-term interest rate	-0.0345	-0.0492	-0.0262
	(0.1414)	(0.1439)	(0.1164)
Rule of law	-0.0373	-0.1161	-0.1152
	(0.1770)	(0.1779)	(0.1652)
Regulatory quality	0.0211	-0.0165	0.0299
	(0.1178)	(0.1211)	(0.0879)
Control of corruption	0.0799	0.0451	0.1536
	(0.1157)	(0.1492)	(0.1193)
Fixed-effects			
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Firms	2,183	1,582	2,165
Years	9	9	9
Observations	18,562	13,587	18,472
Adjusted R^2	0.492	0.481	0.489
Within R^2	0.042	0.026	0.041

Notes: The table reports complete coefficient estimates for the selected robustness specifications. Firm and year fixed effects are included throughout. Standard errors are clustered by headquarters country (20 clusters across the reported models) and appear in parentheses. The dependent variable is adjusted earnings volatility and is expressed in percentage points. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 24: Cash/assets regressions after matching and reweighting

Model:	Main model (1)	PS match (2)	ATT weighted (3)
Variables			
Treat × Post2019	-0.0039	-3.36×10^{-5}	-0.0041
	(0.0068)	(0.0071)	(0.0070)
Log assets	-0.0105***	-0.0126**	-0.0051
	(0.0035)	(0.0054)	(0.0072)
EBIT/assets	0.0439***	0.0868***	0.0491***
	(0.0104)	(0.0139)	(0.0096)
Tangibility	-0.2294***	-0.2228***	-0.2554***
	(0.0233)	(0.0344)	(0.0331)
Debt-to-assets	-0.1193***	-0.1025***	-0.1291***
	(0.0220)	(0.0154)	(0.0186)
Log sales growth	-0.0049***	-0.0055**	-0.0033
	(0.0012)	(0.0022)	(0.0029)
Corporate tax rate	-0.0015***	-0.0016***	-0.0012***
	(0.0004)	(0.0004)	(0.0004)
GDP growth	0.0016***	0.0013***	0.0015***
	(0.0004)	(0.0004)	(0.0005)
Inflation	0.0004	0.0010	0.0004
	(0.0018)	(0.0014)	(0.0016)
Long-term interest rate	0.0032	0.0025*	0.0029
	(0.0020)	(0.0014)	(0.0017)
Rule of law	0.0013	0.0005	0.0014
	(0.0013)	(0.0012)	(0.0012)
Regulatory quality	0.0004	0.0020**	0.0003
	(0.0008)	(0.0009)	(0.0007)
Control of corruption	8.9×10^{-6}	-0.0002	-9.02×10^{-5}
	(0.0007)	(0.0005)	(0.0006)
Fixed-effects			
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Firms	2,427	1,669	2,393
Years	10	10	10
Observations	22,570	15,737	22,355
Adjusted R^2	0.729	0.734	0.735
Within R^2	0.069	0.085	0.078

Notes: The table reports complete coefficient estimates for the selected robustness specifications. Firm and year fixed effects are included throughout. Standard errors are clustered by headquarters country (20 clusters across the reported models) and appear in parentheses. The dependent variable is cash/assets; coefficients are in ratio units. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Matching Pre-Trend Diagnostics

Table 25: Joint pre-trend tests after matching and ATT reweighting

	PS match		ATT	
Outcome	Wald χ^2	p-value	Wald χ^2	p-value
Capital structure				
Debt-to-assets	4.275	0.3700	4.320	0.3645
Log equity	2.076	0.7218	9.989	0.0406
Log current debt	1.954	0.7442	1.266	0.8671
Log long-term debt	45.253	< 0.0001	36.022	< 0.0001
Operational risk				
Adjusted *EarnVol*	13.677	0.0034	9.218	0.0265
Unadjusted *EarnVol*	12.427	0.0061	14.092	0.0028
Cash holdings				
Cash/assets	8.306	0.0810	8.987	0.0614

Notes: The PS design is the selected 1:1 nearest-neighbor match without replacement and with a 0.001 caliper. ATT denotes inverse-propensity reweighting. For capital structure and cash holdings, 2018 is omitted and the tests jointly evaluate the 2014–2017 treated-year coefficients. For operational risk, 2016 is omitted and the tests evaluate the 2013–2015 forward-window anchor years. All models retain the corresponding outcome-specific controls, firm and year fixed effects, and inference clustered by headquarters country. Failure to reject does not establish parallel trends.

COVID-Period Sensitivity

Table 26: Debt-to-assets regressions for COVID-period sensitivity

Model:	Main model (1)	Strict timing (2)	Deaths (3)	Stringency (4)	Joint (5)
Variables					
Treat × Post2019	-0.0205***	-0.0150**	-0.0206***	-0.0186***	-0.0187***
	(0.0068)	(0.0063)	(0.0068)	(0.0064)	(0.0063)
Tangibility	0.2114***	0.1913***	0.2113***	0.2112***	0.2112***
	(0.0326)	(0.0342)	(0.0327)	(0.0327)	(0.0327)
EBITDA/assets	-0.1096***	-0.1033***	-0.1095***	-0.1093***	-0.1093***
	(0.0276)	(0.0283)	(0.0276)	(0.0275)	(0.0275)
Log assets	0.0118	0.0132	0.0118	0.0120	0.0119
	(0.0077)	(0.0078)	(0.0077)	(0.0077)	(0.0077)
Country-sector sales growth	-0.0008	-0.0007	-0.0009	-0.0009	-0.0009
	(0.0030)	(0.0031)	(0.0030)	(0.0030)	(0.0030)
Corporate tax rate	-0.0013**	-0.0010	-0.0013**	-0.0012*	-0.0012*
	(0.0005)	(0.0006)	(0.0006)	(0.0006)	(0.0006)
GDP growth	-0.0009	-0.0006	-0.0009	-0.0008	-0.0008
	(0.0008)	(0.0011)	(0.0008)	(0.0008)	(0.0008)
Inflation	0.0017*	0.0014	0.0015	0.0015	0.0014
	(0.0010)	(0.0009)	(0.0011)	(0.0009)	(0.0010)
Long-term interest rate	0.0017	0.0034**	0.0019	0.0022	0.0023
	(0.0015)	(0.0014)	(0.0015)	(0.0015)	(0.0015)
Rule of law	-0.0031	-0.0043	-0.0031	-0.0030	-0.0030
	(0.0021)	(0.0027)	(0.0022)	(0.0023)	(0.0023)
Regulatory quality	0.0019*	0.0024*	0.0019*	0.0021*	0.0021**
	(0.0011)	(0.0011)	(0.0010)	(0.0010)	(0.0010)
Control of corruption	0.0002	0.0004	0.0001	-0.0003	-0.0003
	(0.0008)	(0.0012)	(0.0008)	(0.0009)	(0.0009)
Log deaths per million			0.0015		0.0004
			(0.0025)		(0.0025)
Oxford stringency / 10				0.0063	0.0061
				(0.0038)	(0.0040)
Fixed-effects					
Firm	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Firms	2,655	2,653	2,655	2,655	2,655
Years	10	8	10	10	10
Observations	24,851	19,576	24,851	24,851	24,851
Adjusted R^2	0.681	0.667	0.681	0.681	0.681
Within R^2	0.041	0.036	0.041	0.041	0.041

Notes: The table reports complete coefficient estimates for the selected robustness specifications. Firm and year fixed effects are included throughout. Standard errors are clustered by headquarters country (20 clusters across the reported models) and appear in parentheses. The strict-timing model excludes 2020–2021. Debt-to-assets coefficients are in ratio units. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 27: Adjusted earnings-volatility regressions for COVID-period sensitivity

Model:	Main model (1)	Strict timing (2)	Deaths (3)	Stringency (4)	Joint (5)
Variables					
Treat × Reform2017	-1.956*	-0.5354	-1.928*	-2.124*	-2.128*
	(1.027)	(0.9156)	(1.018)	(1.140)	(1.141)
Log assets	-3.866***	-3.696***	-3.865***	-3.877***	-3.878***
	(0.6234)	(0.5772)	(0.6229)	(0.6219)	(0.6215)
EBIT/assets	-19.07***	-13.98***	-19.07***	-19.08***	-19.09***
	(2.656)	(3.033)	(2.657)	(2.665)	(2.667)
Log sales growth	-0.7505***	-0.2964	-0.7459***	-0.7475***	-0.7398***
	(0.2554)	(0.3284)	(0.2554)	(0.2572)	(0.2566)
Log sales	3.226***	2.756***	3.226***	3.221***	3.219***
	(0.8083)	(0.7413)	(0.8081)	(0.8063)	(0.8057)
Corporate tax rate	-0.0813	-0.1489	-0.0738	-0.0964	-0.0892
	(0.0940)	(0.0958)	(0.0899)	(0.1066)	(0.1015)
GDP growth	-0.0569	0.1778	-0.0559	-0.1033	-0.1141
	(0.0536)	(0.1514)	(0.0536)	(0.0955)	(0.0993)
Inflation	0.2949	0.1759	0.2369	0.2194	0.1122
	(0.4445)	(0.3351)	(0.4096)	(0.4384)	(0.4055)
Long-term interest rate	-0.0345	0.0707	-0.0206	-0.1395	-0.1464
	(0.1414)	(0.2207)	(0.1455)	(0.2188)	(0.2171)
Rule of law	-0.0373	-0.5799***	-0.0510	-0.0533	-0.0782
	(0.1770)	(0.1817)	(0.1817)	(0.1808)	(0.1905)
Regulatory quality	0.0211	0.2705	0.0173	0.0147	0.0072
	(0.1178)	(0.1658)	(0.1220)	(0.1253)	(0.1327)
Control of corruption	0.0799	0.4091*	0.0758	0.1489	0.1610
	(0.1157)	(0.2161)	(0.1160)	(0.1712)	(0.1761)
Forward log deaths per million			0.4210		0.6328
			(0.5872)		(0.6624)
Forward Oxford stringency / 10				-0.9939	-1.259
				(1.209)	(1.329)
Fixed-effects					
Firm	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Firms	2,183	2,155	2,183	2,183	2,183
Years	9	5	9	9	9
Observations	18,562	9,925	18,562	18,562	18,562
Adjusted R^2	0.492	0.650	0.492	0.492	0.492
Within R^2	0.042	0.030	0.042	0.042	0.042

Notes: The table reports complete coefficient estimates for the selected robustness specifications. Firm and year fixed effects are included throughout. Standard errors are clustered by headquarters country (20 clusters across the reported models) and appear in parentheses. The strict-timing model retains forward volatility windows ending before 2020. The outcome is expressed in percentage points. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 28: Cash/assets regressions for COVID-period sensitivity

Model:	Main model (1)	Strict timing (2)	Deaths (3)	Stringency (4)	Joint (5)
Variables					
Treat × Post2019	-0.0039	-0.0042	-0.0038	-0.0032	-0.0025
	(0.0068)	(0.0058)	(0.0066)	(0.0067)	(0.0063)
Log assets	-0.0105***	-0.0138***	-0.0105***	-0.0105***	-0.0104***
	(0.0035)	(0.0037)	(0.0035)	(0.0036)	(0.0035)
EBIT/assets	0.0439***	0.0434***	0.0437***	0.0440***	0.0438***
	(0.0104)	(0.0104)	(0.0105)	(0.0105)	(0.0105)
Tangibility	-0.2294***	-0.2157***	-0.2293***	-0.2295***	-0.2293***
	(0.0233)	(0.0237)	(0.0233)	(0.0233)	(0.0233)
Debt-to-assets	-0.1193***	-0.1266***	-0.1193***	-0.1194***	-0.1194***
	(0.0220)	(0.0197)	(0.0220)	(0.0220)	(0.0220)
Log sales growth	-0.0049***	-0.0029	-0.0049***	-0.0049***	-0.0049***
	(0.0012)	(0.0019)	(0.0012)	(0.0012)	(0.0012)
Corporate tax rate	-0.0015***	-0.0011***	-0.0016***	-0.0015***	-0.0016***
	(0.0004)	(0.0003)	(0.0004)	(0.0004)	(0.0004)
GDP growth	0.0016***	0.0028***	0.0015***	0.0016***	0.0015***
	(0.0004)	(0.0008)	(0.0004)	(0.0004)	(0.0004)
Inflation	0.0004	0.0003	0.0009	0.0003	0.0008
	(0.0018)	(0.0018)	(0.0017)	(0.0018)	(0.0016)
Long-term interest rate	0.0032	0.0032	0.0029	0.0034	0.0032*
	(0.0020)	(0.0020)	(0.0018)	(0.0020)	(0.0018)
Rule of law	0.0013	0.0020	0.0013	0.0014	0.0013
	(0.0013)	(0.0016)	(0.0011)	(0.0013)	(0.0012)
Regulatory quality	0.0004	0.0002	0.0004	0.0005	0.0006
	(0.0008)	(0.0008)	(0.0008)	(0.0008)	(0.0008)
Control of corruption	8.9×10^{-6}	-0.0004	0.0001	-0.0001	-0.0002
	(0.0007)	(0.0008)	(0.0006)	(0.0007)	(0.0007)
Log deaths per million			-0.0038		-0.0046
			(0.0032)		(0.0029)
Oxford stringency / 10				0.0023	0.0043
				(0.0035)	(0.0034)
Fixed-effects					
Firm	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Firms	2,427	2,426	2,427	2,427	2,427
Years	10	8	10	10	10
Observations	22,570	17,804	22,570	22,570	22,570
Adjusted R^2	0.729	0.722	0.730	0.729	0.730
Within R^2	0.069	0.073	0.070	0.069	0.070

Notes: The table reports complete coefficient estimates for the selected robustness specifications. Firm and year fixed effects are included throughout. Standard errors are clustered by headquarters country (20 clusters across the reported models) and appear in parentheses. The strict-timing model excludes 2020–2021. Cash/assets coefficients are in ratio units. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Pre-Reform Policy Regimes

Table 29: Debt-to-assets regressions by pre-reform policy regime

Model:	No prior rule (1)	Thin-cap only (2)	Non-equivalent ESR (3)
Variables			
Treat × Post2019	-0.0301***	-0.0082	-0.0055
	(0.0026)	(0.0112)	(0.0089)
Tangibility	0.2294***	0.2116***	0.2466***
	(0.0326)	(0.0560)	(0.0411)
EBITDA/assets	-0.0844***	-0.1591***	-0.1305***
	(0.0160)	(0.0399)	(0.0248)
Log assets	0.0102	0.0107	0.0074
	(0.0087)	(0.0176)	(0.0151)
Country-sector sales growth	0.0007	-0.0043	-0.0065*
	(0.0047)	(0.0032)	(0.0030)
Corporate tax rate	-0.0015***	-0.0010	-0.0010**
	(0.0003)	(0.0006)	(0.0004)
GDP growth	-0.0012**	-0.0011	-0.0008
	(0.0005)	(0.0007)	(0.0015)
Inflation	0.0029***	0.0019*	0.0012
	(0.0006)	(0.0010)	(0.0018)
Long-term interest rate	-0.0014	-0.0005	0.0013
	(0.0020)	(0.0021)	(0.0022)
Rule of law	-0.0095***	-0.0060**	-0.0068**
	(0.0019)	(0.0025)	(0.0029)
Regulatory quality	0.0011	0.0014	0.0024
	(0.0007)	(0.0011)	(0.0014)
Control of corruption	-0.0001	-0.0021*	-5.93×10^{-5}
	(0.0008)	(0.0011)	(0.0010)
Fixed-effects			
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Firms	1,923	1,375	1,569
Years	10	10	10
Observations	18,009	13,089	14,789
Adjusted R^2	0.692	0.719	0.693
Within R^2	0.044	0.054	0.050

Notes: The table reports complete coefficient estimates for the selected robustness specifications. Firm and year fixed effects are included throughout. Standard errors are clustered by headquarters country (10–12 clusters across the reported models) and appear in parentheses. Each specification retains the complete control group and one treated-country regime. Debt-to-assets coefficients are in ratio units. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 30: Adjusted earnings-volatility regressions by pre-reform policy regime

Model:	No prior rule (1)	Thin-cap only (2)	Non-equivalent ESR (3)
Variables			
Treat × Reform2017	-3.772*	0.3279	-0.9020
	(1.811)	(0.5012)	(1.418)
Log assets	-3.535***	-4.350***	-4.709***
	(0.6275)	(1.023)	(0.8638)
EBIT/assets	-17.47***	-17.64***	-19.02***
	(2.157)	(3.675)	(4.666)
Log sales growth	-0.5178**	-0.6586*	-1.027**
	(0.1644)	(0.3421)	(0.3502)
Log sales	2.898***	2.907***	4.009***
	(0.8393)	(0.6983)	(0.8700)
Corporate tax rate	-0.0312	0.0289	0.0122
	(0.0719)	(0.0654)	(0.0792)
GDP growth	-0.1162	0.0171	-0.0385
	(0.1082)	(0.1020)	(0.1047)
Inflation	0.1230	0.7105*	-0.0229
	(0.4728)	(0.3263)	(0.6737)
Long-term interest rate	-0.0206	0.1568	-0.0310
	(0.1707)	(0.1188)	(0.1320)
Rule of law	-0.3715	-0.3810*	-0.2360
	(0.2750)	(0.1824)	(0.1958)
Regulatory quality	-0.0998	2.88×10^{-5}	0.0583
	(0.1708)	(0.1046)	(0.1231)
Control of corruption	0.1763	0.3315**	0.1677
	(0.1963)	(0.1295)	(0.1696)
Fixed-effects			
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Firms	1,568	1,201	1,328
Years	9	9	9
Observations	13,286	10,377	11,459
Adjusted R^2	0.466	0.524	0.401
Within R^2	0.039	0.038	0.039

Notes: The table reports complete coefficient estimates for the selected robustness specifications. Firm and year fixed effects are included throughout. Standard errors are clustered by headquarters country (10–12 clusters across the reported models) and appear in parentheses. Each specification retains the complete control group and one treated-country regime. The outcome is expressed in percentage points. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 31: Cash/assets regressions by pre-reform policy regime

Model:	No prior rule (1)	Thin-cap only (2)	Non-equivalent ESR (3)
Variables			
Treat × Post2019	-0.0017	0.0031	0.0018
	(0.0075)	(0.0066)	(0.0066)
Log assets	-0.0108**	-0.0006	-0.0096
	(0.0041)	(0.0072)	(0.0069)
EBIT/assets	0.0491***	0.0636**	0.0629***
	(0.0142)	(0.0209)	(0.0157)
Tangibility	-0.2553***	-0.2320***	-0.2469***
	(0.0253)	(0.0384)	(0.0362)
Debt-to-assets	-0.1173***	-0.0876***	-0.1022***
	(0.0285)	(0.0208)	(0.0218)
Log sales growth	-0.0041***	-0.0040**	-0.0057*
	(0.0012)	(0.0018)	(0.0026)
Corporate tax rate	-0.0013**	-0.0013***	-0.0014***
	(0.0004)	(0.0004)	(0.0004)
GDP growth	0.0019***	0.0011*	0.0012***
	(0.0004)	(0.0005)	(0.0003)
Inflation	-0.0029*	0.0034**	9.7×10^{-5}
	(0.0014)	(0.0014)	(0.0007)
Long-term interest rate	0.0028**	0.0014	0.0004
	(0.0010)	(0.0010)	(0.0008)
Rule of law	0.0016	-0.0017	-0.0005
	(0.0012)	(0.0010)	(0.0008)
Regulatory quality	0.0012	0.0017	0.0003
	(0.0010)	(0.0010)	(0.0011)
Control of corruption	-0.0008	0.0007	-0.0002
	(0.0011)	(0.0009)	(0.0006)
Fixed-effects			
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Firms	1,746	1,291	1,460
Years	10	10	10
Observations	16,189	12,245	13,750
Adjusted R^2	0.731	0.736	0.749
Within R^2	0.072	0.080	0.087

Notes: The table reports complete coefficient estimates for the selected robustness specifications. Firm and year fixed effects are included throughout. Standard errors are clustered by headquarters country (10–12 clusters across the reported models) and appear in parentheses. Each specification retains the complete control group and one treated-country regime. Cash/assets coefficients are in ratio units. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Firm-Level Heterogeneity

Table 32: Debt-to-assets regressions by pre-reform exposure proxy

Model:	Leverage by risk quartile (1)	Leverage by accounting pressure (2)
Interaction terms		
Treat × Post2019	0.0031	-0.0188**
	(0.0125)	(0.0081)
Q2 × Post2019	0.0181**	
	(0.0085)	
Q3 × Post2019	0.0133	
	(0.0176)	
Q4 × Post2019	0.0325***	
	(0.0082)	
Treat × Post2019 × Q2	-0.0164	
	(0.0145)	
Treat × Post2019 × Q3	-0.0201	
	(0.0198)	
Treat × Post2019 × Q4	-0.0337***	
	(0.0109)	
High pressure × Post2019		-0.0419***
		(0.0076)
Treat × Post2019 × High pressure		0.0042
		(0.0094)
Firm and country-year controls		
Tangibility	0.2056***	0.2244***
	(0.0415)	(0.0353)
EBITDA/assets	-0.1419***	-0.1247***
	(0.0303)	(0.0308)
Log assets	0.0257***	0.0235***
	(0.0079)	(0.0069)
Country-sector sales growth	0.0001	0.0010
	(0.0036)	(0.0032)
Corporate tax rate	-0.0010*	-0.0017**
	(0.0006)	(0.0006)
GDP growth	-0.0011	-0.0010
	(0.0008)	(0.0007)
Inflation	0.0012	0.0014
	(0.0010)	(0.0010)
Long-term interest rate	0.0012	0.0014
	(0.0016)	(0.0020)
Rule of law	-0.0041*	-0.0031
	(0.0022)	(0.0023)
Regulatory quality	0.0032**	0.0029**
	(0.0012)	(0.0013)
Control of corruption	-0.0004	-0.0006
	(0.0009)	(0.0010)
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Firms	1,958	1,974
Observations	19,431	19,493
Countries	20	20
Adjusted R^2	0.702	0.702
Within R^2	0.057	0.066

Notes: The dependent variable is debt-to-assets; coefficients are in ratio units. Column (1) uses Q1 as the omitted operational-risk group. Column (2) uses low accounting pressure as the omitted group. Operational risk and accounting pressure are defined in the notes to Table 10. Both specifications include the full leverage control set, firm fixed effects, and year fixed effects. Standard errors are clustered by headquarters country and appear in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

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